**Group 13: Boron Family ($ns^2\ np^1$)** — Periodic Trends & Anomalies

Group 13 contains B (non-metal), Al (metal/amphoteric), Ga, In, Tl (post-transition metals). The chemistry is dominated by: small size of B, inert pair effect descending the group, and poor d/f electron shielding causing atomic radius and ionization enthalpy anomalies.

## 1. Atomic Radii & Ionization Enthalpy Anomalies

| Element | IE-1 (kJ/mol) | Atomic Radius (pm) | Key Electronic Reason |
|---|---|---|---|
| B [He] $2s^2\ 2p^1$ | 801 | 85 | Extremely small size, covalent network, high charge density |
| Al [Ne] $3s^2\ 3p^1$ | 577 | 143 | Normal expansion with full 8-electron noble core |
| Ga [Ar] $3d^{10}\ 4s^2\ 4p^1$ | 579 | 135 | r(Ga) < r(Al)! Poor shielding by $3d^{10}$ → $Z_{eff}$ increases |
| In [Kr] $4d^{10}\ 5s^2\ 5p^1$ | 558 | 167 | Normal size increase after 4d shell |
| Tl [Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^1$ | 589 | 170 | IE(Tl) > IE(In)! Lanthanoid contraction (poor 4f shielding) |

Table 13.1 — G13 physical property anomalies (NCERT & JD Lee).

KEY RESULT
Atomic Radius Order: **B < Ga < Al < In ≈ Tl** (Ga smaller than Al due to 3d contraction).
Ionization Enthalpy Order: **B > Tl > Ga > Al > In** (Tl has 2nd highest IE due to 4f contraction).

## 2. Inert Pair Effect & Oxidation States (+3 vs +1)

Descending the group, the $6s^2$ pair of valence electrons becomes increasingly reluctant to participate in bonding due to high effective nuclear charge following 4f and 5d subshells.

- **+3 Stability Order:** B(+3) > Al(+3) > Ga(+3) > In(+3) > Tl(+3). $Tl^{3+}$ is a powerful **oxidising agent** ($Tl^{3+} + 2e^- \rightarrow Tl^+$, $E^\circ = +1.26$ V).
- **+1 Stability Order:** B(+1) < Al(+1) < Ga(+1) < In(+1) < **Tl(+1)**. TlCl, TlOH are ionic and stable like alkali metals (TlOH is a strong base).
- **Al(+1) and Ga(+1):** Disproportionate in water: $3\ Ga^+ \rightarrow 2\ Ga + Ga^{3+}$.

TRAP
TRAP: TlI3 is NOT Thallium(III) iodide! It is an ionic compound **$Tl^+$ ($I_3^-$)** containing the linear triiodide anion. $Tl^{3+}$ readily oxidises $I^-$ to $I_2$!

## 3. Extraction & Reactivity of Boron

Boron is extracted from **Borax ($Na_2B_4O_7{\cdot}10H_2O$)** or **Colemanite ($Ca_2B_6O_{11}{\cdot}5H_2O$)**:
1. Boil ore with dil. $H_2SO_4$ → precipitate orthoboric acid $H_3BO_3$.
2. Ignite $H_3BO_3 \rightarrow B_2O_3$. Then reduce: $B_2O_3 + 3\ Mg \rightarrow 2\ B$ (amorphous) $+ 3\ MgO$.
3. Ultra-pure crystalline B via van Arkel: 2 BBr3 + 3 $H_2$ (Ta filament, 1200°C) → 2 B + 6 HBr.

**Diborane ($B_2H_6$) — Synthesis, 3c-2e Bonding, Cleavage & Reactions**

Boron has only 3 valence electrons, so monomeric BH3 has only 6 valence electrons (Lewis acid). It dimerises to $B_2H_6$ via two electron-deficient bridging bonds (multicentre 3c-2e bonds).

## 1. Structure & Multi-Centre Bonding of B2H6

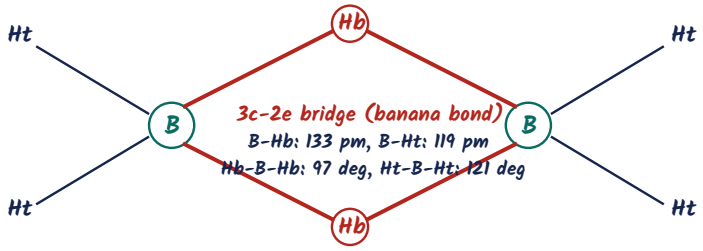



Fig. 13.1 — Diborane geometry with terminal (Ht) and bridging (Hb) hydrogens.

- **Terminal B-Ht:** 4 normal covalent 2c-2e bonds, coplanar in the xy-plane (B-Ht = 119 pm, H-B-H = 121°).
- **Bridging B-Hb-B:** 2 three-centre two-electron (3c-2e) 'banana' bonds, lying above and below the xy-plane (B-Hb = 133 pm, B-Hb-B = 97°).
- **Hybridization of B:** Each Boron atom is **$sp^3$ hybridized** (4 electron domains). Total valence electrons = 2×3 + 6×1 = 12 electrons (6 pairs for 8 bond vectors).

## 2. Cleavage of B2H6: Symmetrical vs Unsymmetrical

The mode of Lewis base cleavage depends directly on the base's steric size and hardness:

| Type of Cleavage | Reagents (Lewis Bases) | Product & Reaction Equation |
|---|---|---|
| Symmetrical Cleavage | Bulky/Soft bases: NMe3, THF, CO, SMe2, PMe3 | $B_2H_6$ + 2 NMe3 → 2 $Me_3N{:}BH_3$ (splits into two BH3·L) |
| Unsymmetrical Cleavage | Small/Hard bases: NH3 (low T), MeNH2, $Me_2NH$ | $B_2H_6$ + 2 NH3 → $[H_2B(NH_3)_2]^+$ $[BH_4]^-$ (ionic salt) |

Table 13.2 — Cleavage mechanisms of Diborane (NCERT & JD Lee core).

KEY RESULT
Industrial Prep: 2 BF3 + 6 NaH → $B_2H_6$ + 6 NaF (at 450 K in polyether).
Laboratory Prep: 4 BF3·OEt2 + 3 LiAlH4 → 2 $B_2H_6$ + 3 LiAlF4 + 4 $Et_2O$.
Redox Prep: 2 NaBH4 + $I_2$ → $B_2H_6$ + 2 NaI + $H_2$ (in diglyme).

## 3. Reactions of B2H6

- **Combustion:** $B_2H_6$ + 3 $O_2$ → $B_2O_3$ + 3 $H_2O$ ($\Delta H°$ = -1976 kJ/mol, violent green flame, high fuel value).
- **Hydrolysis:** $B_2H_6$ + 6 $H_2O$ → 2 $H_3BO_3$ + 6 $H_2$ (rapid, quantitative evolution of hydrogen).
- **Hydroboration:** $B_2H_6$ + 6 RCH=CH2 → 2 $B(CH_2CH_2R)_3$ (syn-addition, anti-Markovnikov alcohol precursor).

**Higher Boranes & Hydroborates** — Wade's Rules, $NaBH_4$ & Borazine

Boranes form extensive cluster families classified by skeletal electron pairs (Wade-Mingos rules): Closo ($B_n H_{n+2}$)$^{2-}$, Nido ($B_n H_{n+4}$), Arachno ($B_n H_{n+6}$), and Hypho ($B_n H_{n+8}$).

## 1. Borazine $B_3N_3H_6$ ('Inorganic Benzene')

Heating the unsymmetrical cleavage product of diborane and ammonia gives Borazine:
$3\ B_2H_6 + 6\ NH_3$ (low T) $\rightarrow 3\ [H_2B(NH_3)_2]^+[BH_4]^-$ —(heat, 450 K)→ **$2\ B_3N_3H_6 + 12\ H_2$**
Alternate synthesis: $3\ BCl_3 + 3\ NH_4Cl \rightarrow (ClBNH)_3 + 9\ HCl$; reduce with $NaBH_4 \rightarrow B_3N_3H_6$.

| Property | Benzene ($C_6H_6$) | Borazine ($B_3N_3H_6$) |
|---|---|---|
| Aromaticity | Completely delocalised (150 kJ/mol) | Partially delocalised (pi polar towards N) |
| Polarity & Reactivity | Non-polar, electrophilic substitution | Polar ($B^{\delta+}$ - $^{N}\delta-$), addition reactions dominant |
| Addition of HCl | No reaction without $AlCl_3$ | Adds 3 HCl readily: H to N, Cl to B → $B_3N_3H_9Cl_3$ |
| Hydrolysis | Resistant to hydrolysis | Hydrolyses readily: $B_3N_3H_6 + 9\ H_2O \rightarrow 3\ H_3BO_3 + 3\ NH_3 + 3\ H_2$ |

Table 13.3 — Comparison of Benzene vs Inorganic Benzene.

KEY RESULT
Borazine is **iso-electronic and iso-structural** to Benzene (planar hexagonal ring, $sp^2$ B and N), but chemically MUCH more reactive due to electronegativity difference ($\chi_N = 3.0$, $\chi_B = 2.0$).

## 2. Sodium Borohydride ($NaBH_4$) & $LiAlH_4$

- **$NaBH_4$ Synthesis:** $4\ NaH + B(OMe)_3 \rightarrow NaBH_4 + 3\ NaOMe$ (at 250°C).
  Structure: Ionic $[Na]^+\ [BH_4]^-$ (tetrahedral $BH_4^-$, $sp^3$ B, regular 2c-2e B-H bonds, 8 valence $e^-$).
- **$LiAlH_4$ Synthesis:** $4\ LiH + AlCl_3 \rightarrow LiAlH_4 + 3\ LiCl$ (in dry ether).
  Reducing power: $LiAlH_4$ is a far stronger hydride donor than $NaBH_4$ because Al-H is more polar and weaker than B-H. $LiAlH_4$ reacts violently with water/alcohols; $NaBH_4$ can be used in basic water/MeOH.

TRAP
TRAP: $NaBH_4$ reduces aldehydes and ketones to alcohols, but does NOT reduce esters, amides, or carboxylic acids under standard conditions. $LiAlH_4$ reduces all of them.

**Boron Trihalides (BX3)** — pπ-pπ Back-Bonding & Lewis Acidity Inversion

In BX3 (X = F, Cl, Br, I), Boron is $sp^2$ hybridized with a vacant, unhybridized $2p_z$ orbital perpendicular to the planar molecule. The halogen atom has filled p orbitals with lone pairs.

## 1. pπ-pπ Back-Bonding Mechanism

The lone pair in a filled 2p orbital of Fluorine overlaps sideways with the empty 2p orbital of Boron, forming a **pπ-pπ dative back-bond**. This gives partial double bond character to the B-X bond.

| Halide | Back-Bond Overlap | Extent of Back-Bonding | Observed B-X Bond Length |
|---|---|---|---|
| BF3 | 2p(B) ← 2p(F) (identical size, excellent match) | Maximum back-bonding | 130 pm (much shorter than single bond 152 pm) |
| BCl3 | 2p(B) ← 3p(Cl) (diffuse 3p, poor overlap) | Weak back-bonding | 175 pm |
| BBr3 | 2p(B) ← 4p(Br) (very diffuse 4p) | Very weak back-bonding | 187 pm |
| BI3 | 2p(B) ← 5p(I) (energy/size mismatch) | Negligible back-bonding | 210 pm |

Table 13.4 — Back-bonding variation across Boron Trihalides.

KEY RESULT
LEWIS ACIDITY ORDER: **BF3 < BCl3 < BBr3 < BI3** (Opposite of halogen electronegativity!).
Reason: In BF3, strong 2p-2p back-bonding satisfies Boron's electron deficiency. In BI3, negligible back-bonding leaves Boron completely vacant and most hungry for external electron pairs.

## 2. Hydrolysis of Boron Trihalides

- **BCl3, BBr3, BI3:** Undergo complete hydrolysis without secondary complexation:
  $BX_3 + 3\ H_2O \rightarrow H_3BO_3 + 3\ HX$ (vigorous, fumes in moist air).
- **BF3 Anomaly (Partial Hydrolysis):**
  Step 1: $4\ BF_3 + 3\ H_2O \rightarrow B(OH)_3 + 3\ HBF_4$ (or: $BF_3 + 3\ H_2O \rightarrow H_3BO_3 + 3\ HF$)
  Step 2: HF immediately attacks unreacted BF3: $3\ BF_3 + 3\ HF \rightarrow 3\ H^+\ [BF_4]^-$ (Fluoroboric acid).
  Net reaction: $\mathbf{4\ BF_3 + 3\ H_2O \rightarrow H_3BO_3 + 3\ H[BF_4]}$ (does not go to completion to yield pure HF!).

TRAP
TRAP: BF3 does NOT give white fumes of HF on hydrolysis because HF reacts with BF3 to form HBF4! BCl3 fumes copiously because HCl gas does not coordinate to BCl3.

**Orthoboric Acid & Borax** — Layered Hydrogen Bonding & Aprotic Acidity

Orthoboric acid $H_3BO_3$ (or $B(OH)_3$) is a white, slippery, crystalline solid with a layered sheet structure held together by extensive planar hydrogen bonds. It acts as a mild antiseptic.

## 1. Nature of Acidity: Monobasic Lewis Acid

$H_3BO_3$ is **NOT a protonic Bronsted acid** (does not lose its own $H^+$). It acts as a **monobasic Lewis acid** by accepting $OH^-$ from water, thereby releasing $H^+$ from the solvent:

KEY RESULT
$B(OH)_3 + 2\ H_2O \rightleftharpoons [B(OH)_4]^- + H_3O^+$ ($pK_a = 9.25$, very weak)
It cannot be titrated directly against standard NaOH because the endpoint is diffuse.

## 2. Enhancement of Acidity by Polyhydroxy Compounds (cis-Diols)

When cis-diols (glycerol, mannitol, catechol, or sorbitol) are added, the $[B(OH)_4]^-$ chelate forms a very stable spirocyclic bis-chelate complex, shifting the equilibrium completely to the right:

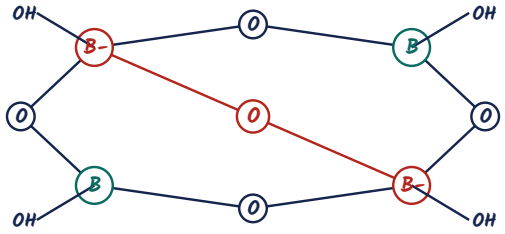
- Reaction: $B(OH)_3 + 2\ C_3H_5(OH)_3 \rightleftharpoons [B(C_3H_6O_3)_2]^- + H_3O^+ + 2\ H_2O$.
- Result: Boric acid becomes a **strong monoprotic acid** and can be titrated sharply against NaOH using phenolphthalein indicator!
- Requirement: The diol must have **cis-vicinal -OH groups** (trans-diols do not enhance acidity).

## 3. Thermal Action on Boric Acid

Stepwise dehydration on heating:
$H_3BO_3$ —(373 K, $-H_2O$)→ **HBO2 (Metaboric acid)** —(433 K, $-H_2O$)→ **$H_2B_4O_7$ (Tetraboric acid)** —(red heat)→ **$B_2O_3$ (Boron sesquioxide / glassy mass)**.

## 4. Borax (Na2B4O7·10H2O)

Correct structural formula: **$Na_2[B_4O_5(OH)_4]\cdot 8H_2O$**. It contains 8 water molecules of crystallization.




[B4O5(OH)4]2- : Two sp2 B (planar) + Two sp3 B- (tetrahedral) + 5 B-O-B bridges

Fig. 13.2 — Structure of the $[B_4O_5(OH)_4]2-$ anion in Borax.

- **Borax in Water:** Undergoes hydrolysis to give an alkaline buffer solution:
  $Na_2B_4O_7 + 7\ H_2O \rightarrow 2\ NaOH + 4\ H_3BO_3$ (solution is alkaline, pH ≈ 9).

**Borax Bead Test & Chemical Analysis** — Colored Metaborates & Fluoroborates

On heating in a platinum wire loop, borax swells up as water of crystallization is lost, and then melts into a clear, transparent, glassy bead containing **$NaBO_2 + B_2O_3$**.

## 1. Chemistry of the Borax Bead Test

$Na_2B_4O_7{\cdot}10H_2O \xrightarrow{\text{(heat, -10 } H_2O)} Na_2B_4O_7 \xrightarrow{\text{(fuse)}}$ **2 NaBO2 (Sodium metaborate) + $B_2O_3$ (Boric anhydride)**
When heated with transition metal salts, $B_2O_3$ reacts with metal oxides to form characteristically coloured **metaborates $M(BO_2)_2$**:

| Metal Cation | Oxidising Flame (Hot / Cold) | Reducing Flame (Hot / Cold) | Metaborate Formed |
|---|---|---|---|
| $Cu^{2+}$ | Green (hot) / Blue (cold) | Colorless (hot) / Opaque Red (cold, Cu metal) | $Cu(BO_2)_2$ / Cu |
| $Co^{2+}$ | Deep Blue (hot & cold) | Deep Blue (hot & cold) | $Co(BO_2)_2$ |
| $Cr^{3+}$ | Dark Yellow (hot) / Green (cold) | Green (hot & cold) | $Cr(BO_2)_3$ |
| $Fe^{3+}$ | Yellowish-brown (cold) | Bottle Green (cold) | $Fe(BO_2)_2$ / $Fe(BO_2)_3$ |
| $Ni^{2+}$ | Violet/Brown (hot) / Brown (cold) | Grey (reduced Ni particles) | $Ni(BO_2)_2$ |

Table 13.5 — Characteristic flame colours in Borax Bead Test.

## 2. Qualitative Test for Borate: Ethyl Borate Test

Borate salt + conc. $H_2SO_4$ + Ethanol → Heat → **Triethyl borate $B(OC_2H_5)_3$** vapours burn with a characteristic **green-edged flame**:
$H_3BO_3 + 3\ C_2H_5OH \xrightarrow{\text{(conc. } H_2SO_4)} B(OC_2H_5)_3\uparrow + 3\ H_2O$.

## 3. Fluoroboric Acid ($HBF_4$)

- Preparation: $B(OH)_3 + 4\ HF \rightarrow H[BF_4] + 3\ H_2O$.
- Structure: $[BF_4]^-$ is perfectly tetrahedral ($sp^3$, 4 equivalent B-F bonds of 143 pm).
- Properties: Strong acid, non-coordinating anion, used in Balz-Schiemann diazotisation ($ArN_2^+\ BF_4^- \rightarrow ArF + N_2 + BF_3$).

**Boron Nitride $(BN)_x$ — Synthesis, Allotropes & Comparison with Carbon**

Boron Nitride is iso-electronic with Carbon (B + N = 3 + 5 = 8 $e^-$ = 2 × C). It exists in two major crystalline forms: Hexagonal BN (h-BN, inorganic graphite) and Cubic BN (c-BN, borazon, inorganic diamond).

## 1. Hexagonal Boron Nitride (h-BN: 'Inorganic Graphite')

- **Synthesis:** $B_2O_3$ + 2 NH3 —(1200 K)→ 2 BN + 3 $H_2O$ (or: BCl3 + NH3 → BN + 3 HCl at 1000°C).
- **Structure:** Planar hexagonal rings of alternating B and N atoms (B-N bond length 145 pm).
- **Layer Stacking Difference:** In graphite, layers are stacked in an ABAB pattern with atoms displaced. In h-BN, layers are stacked directly **AA'AA'** with B atoms directly over N atoms in adjacent layers (dipole-dipole attraction $B^{\delta+}\cdots N\delta-$).
- **Electrical Conductivity:** Graphite is a good conductor (delocalised π electrons). h-BN is a **white insulator** (wide bandgap ≈ 5.9 eV) because π electrons are tightly held by electronegative Nitrogen atoms!

| Feature | Graphite (Carbon) | Hexagonal Boron Nitride (h-BN) |
|---|---|---|
| Color & Appearance | Grey-black, metallic luster | Pure white solid ('White Graphite') |
| Layer Stacking | ABAB... (staggered rings) | AA'AA'... (B above N eclipsed) |
| Electrical Conduction | Conductor along planes (semimetal) | Electrical Insulator (wide bandgap) |
| Chemical Resistance | Oxidises at 700°C | Resistant to oxidation up to 1000°C, inert to acids |

Table 13.6 — Direct structural comparison between Graphite and h-BN.

## 2. Cubic Boron Nitride (c-BN: 'Borazon')

Formed by heating h-BN under extreme pressure (60,000 atm, 1800°C). It has a zinc-blende (diamond-like) three-dimensional tetrahedral network. It is second only to diamond in hardness, but has higher thermal stability in air and does not react with ferrous metals during high-speed cutting.

**Aluminium Chemistry** — Halide Dimers, Alums, Passivation & Thermite

Aluminium is the most abundant metal in the Earth's crust (8.3% by mass). Key ore is **Bauxite** **AlOx(OH)3-2x** purified by Bayer's process and reduced by the Hall-Héroult electrolytic process.

## 1. Aluminium Chloride (AlCl3 vs Al2Cl6)

- **Anhydrous AlCl3:** White crystalline solid, highly hygroscopic. Prepared by: $2\ Al + 3\ Cl_2 \rightarrow 2\ AlCl3$ (or: $Al_2O_3 + 3\ C + 3\ Cl_2 \rightarrow 2\ AlCl3 + 3\ CO$).
- **Dimerisation in Vapor/Non-polar solvents:** AlCl3 dimerises to **$Al_2Cl_6$** through two bridging dative bonds ($Cl: \rightarrow Al$). Each Al is $sp^3$ hybridized with an octet.
- **High Temperature Dissociation:** At T > 1000 K, $Al_2Cl_6$ dissociates into planar monomeric **AlCl3** ($sp^2$, Lewis acid).
- **In Water:** $Al_2Cl_6$ dissolves exothermically and ionizes: $Al_2Cl_6 + 12\ H_2O \rightarrow 2\ [Al(H_2O)_6]^{3+} + 6\ Cl^-$ (octahedral complex, acidic pH due to hydrolysis).

KEY RESULT
AlCl3 cannot be prepared anhydrous by heating AlCl3·6H2O! On heating, it undergoes hydrolysis: $2\ AlCl3{\cdot}6H2O \xrightarrow{(heat)} Al_2O_3 + 6\ HCl + 9\ H_2O$.

## 2. Alums: General Formula M2SO4·M'2(SO4)3·24H2O

Where M = univalent cation ($K^+$, $Na^+$, $NH4^+$, $Rb^+$, $Cs^+$; NOT $Li^+$ due to small size) and M' = trivalent cation ($Al^{3+}$, $Cr^{3+}$, $Fe^{3+}$, $Co^{3+}$).
Structural formula: **$[M(H_2O)_6]^+\ [M'(H_2O)_6]^{3+}\ (SO4^{2-})2$**.

| Common Alum | Formula | Primary Uses |
|---|---|---|
| Potash Alum | $K_2SO_4{\cdot}Al_2(SO_4)_3{\cdot}24H_2O$ | Water purification (coagulation of colloidal clay), mordant in dyeing |
| Ammonium Alum | $(NH4)2SO4{\cdot}Al_2(SO_4)_3{\cdot}24H_2O$ | Baking powder, water treatment, fire-retardant textiles |
| Chrome Alum | $K_2SO_4{\cdot}Cr_2(SO_4)_3{\cdot}24H_2O$ | Leather tanning (cross-links collagen proteins) |
| Ferric Alum | $(NH4)2SO4{\cdot}Fe_2(SO_4)_3{\cdot}24H_2O$ | Indicator in Volhard's titration of halides |

Table 13.7 — Composition and applications of Alums in JEE.

## 3. Passivation & Goldschmidt Thermite Process

- **Passivation:** Conc. HNO3 makes Al passive by forming a dense, insoluble, protective oxide film of $Al_2O_3$. Hence, conc. HNO3 can be transported in Al containers.
- **Thermite Reaction:** $Fe_2O_3 + 2\ Al \rightarrow Al_2O_3 + 2\ Fe(l)$ ($\Delta H$ = -850 kJ/mol, T ≈ 3000°C, used for on-site welding of railway tracks).

**Group 14: Carbon Family ($ns^2\ np^2$)** — Periodic Trends, Catenation & Inert Pair

Contains C (non-metal), Si (metalloid), Ge (metalloid), Sn and Pb (metals). Key characteristics: remarkable catenation ability of Carbon, pπ-pπ vs pπ-dπ multiple bonding, and strong inert pair effect making Pb(II) far more stable than Pb(IV).

## 1. Periodic Trends in Group 14

| Element | Config. | Covalent Radius (pm) | IE-1 (kJ/mol) | M.P. (°C) | Electronegativity |
|---|---|---|---|---|---|
| C | [He] $2s^2\ 2p^2$ | 77 | 1086 | 4100 (diamond) | 2.5 |
| Si | [Ne] $3s^2\ 3p^2$ | 118 | 786 | 1410 | 1.8 |
| Ge | [Ar] $3d^{10}\ 4s^2\ 4p^2$ | 122 | 761 | 937 | 1.8 |
| Sn | [Kr] $4d^{10}\ 5s^2\ 5p^2$ | 140 | 708 | 232 | 1.8 |
| Pb | [Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^2$ | 146 | 715 (Pb > Sn!) | 327 | 1.9 |

Table 14.1 — Periodic properties of Group 14 elements.

KEY RESULT
Ionization Enthalpy: **C > Si > Ge > Pb > Sn** (Pb has higher IE than Sn due to poor 4f shielding). Catenation Order: **C >>> Si > Ge ≈ Sn >> Pb** (Directly follows E-E single bond strength: C-C 348 > Si-Si 297 > Ge-Ge 260 > Sn-Sn 240 kJ/mol).

## 2. Inert Pair Effect & Redox Behaviour of Sn and Pb

- **+4 Oxidation State:** C(+4), Si(+4), Ge(+4), Sn(+4) are stable. Pb(+4) is unstable and a **powerful oxidising agent** ($PbO_2 + 4\ HCl \rightarrow PbCl_2 + Cl_2 + 2\ H_2O$).
- **+2 Oxidation State:** Sn(+2) is a **strong reducing agent** ($Sn^{2+} \rightarrow Sn^{4+} + 2e^-$, $E^\circ = -0.15$ V). Used to reduce $FeCl_3$ to $FeCl_2$, and $HgCl_2$ to white $Hg_2Cl_2$ then black Hg.
- **Pb(+2) vs Pb(+4):** Pb(+2) is extremely stable. $PbCl_4$ decomposes on warming to $PbCl_2 + Cl_2$. $PbI_4$ and $PbBr_4$ do not even exist because $I^-$ and $Br^-$ reduce $Pb^{4+}$ instantly!

TRAP
TRAP: $PbO_2$ dissolves in conc. $HNO_3$ with evolution of $O_2$: $2\ PbO_2 + 4\ HNO_3 \rightarrow 2\ Pb(NO_3)_2 + 2\ H_2O + O_2\uparrow$. It does NOT form a nitrate of lead(IV)!

## 3. Multiple Bonding: pπ-pπ vs dπ-pπ

Carbon forms very strong **pπ-pπ multiple bonds** (C=C, C≡C, C=O, C≡N) because 2p orbitals are compact and overlap sideways efficiently. Heavier elements (Si, Ge, Sn) have large, diffuse p orbitals that cannot form stable pπ-pπ bonds; they prefer single bonds or use empty d orbitals (dπ-pπ).

**Allotropes of Carbon** — Diamond, Graphite, Fullerenes & Thermodynamic Stability

Carbon exhibits rich allotropy due to catenation and ability to adopt $sp^3$, $sp^2$, and sp hybridization.

## 1. Structural Comparison of Major Allotropes

| Property | Diamond | Graphite | Fullerene ($C_{60}$) | |
|---|---|---|---|---|
| Hybridization & Geometry | $sp^3$, 3D tetrahedral network | $sp^2$, 2D planar hexagonal sheets | $sp^2$ (curved), soccer-ball ($I_h$ symmetry) | |
| C-C Bond Length | 154 pm (all uniform single bonds) | 141.5 pm (in-plane), 340 pm (inter-layer) | 145 pm (single, 6-5 ring), 138 pm (double, 6-6 ring) | |
| Conductivity | Electrical insulator (bandgap 5.5 eV) | Good electrical conductor along planes | Semiconductor; $K_3C_{60}$ is a superconductor (18 K) | |
| Density & Hardness | 3.51 g/cm³, Hardest natural substance | 2.22 g/cm³, Soft, slippery (solid lubricant) | 1.65 g/cm³, Molecular solid (soluble in toluene) | |
| Standard Heat of Formation | +1.90 kJ/mol (metastable at RTP) | **0.00 kJ/mol (Thermodynamically MOST stable)** | +38.1 kJ/mol | |

Table 14.2 — Comparative crystallography and physics of Carbon allotropes.

KEY RESULT

**Thermodynamic Stability:** Graphite is the standard state of carbon ($\Delta H^\circ_f = 0$). Diamond conversion to graphite is thermodynamically favored ($\Delta G < 0$) but kinetically dormant (huge activation energy).

## 2. Buckminsterfullerene (C60) Architecture

- **Faces & Rings:** $C_{60}$ contains **20 six-membered rings (hexagons)** and **12 five-membered rings (pentagons)**.
- **Isolated Pentagon Rule (IPR):** Six-membered rings can fuse with both hexagons and pentagons, but **five-membered rings ONLY fuse with six-membered rings** (no two pentagons share an edge, minimising strain).
- **Bond Alternation:** C=C double bonds exist strictly at the junction of two hexagons (6-6 ring fusion, length 138 pm). Single bonds exist at 6-5 junctions (145 pm).

## 3. Amorphous Carbon Allotropes & Adsorption

Coke, Charcoal, Lampblack, Carbon black, and Activated Charcoal. Activated charcoal has an immense surface area (≈1000 $m^2/g$) and is used in gas masks, decolourisation of sugar, and water filtration.

**Carbides** — Ionic (Salt-like), Covalent, Interstitial & Hydrolysis Pathways

Carbides are binary compounds of carbon with elements of lower or comparable electronegativity. They are classified into three distinct structural classes based on bonding.

## 1. Salt-like (Ionic) Carbides & Hydrolysis Products

Formed by highly electropositive metals (Group 1, 2, Al). Classified by the hydrocarbon gas evolved on hydrolysis:

| Carbide Type | Carbon Anion | Example Carbides | Hydrolysis Reaction & Gas Evolved |
|---|---|---|---|
| Methanides | $C^{4-}$ (isolated carbon) | $Al_4C_3$, $Be_2C$ | $Al_4C_3 + 12\ H_2O \rightarrow 4\ Al(OH)_3 + \mathbf{3\ CH_4}\uparrow$ (Methane) |
| Acetylides | $C_2^{2-}$ ($[:C\equiv C:]^{2-}$) | CaC2, MgC2, $Cu_2C_2$, $Ag_2C_2$ | $CaC2 + 2\ H_2O \rightarrow Ca(OH)_2 + \mathbf{C_2H_2}\uparrow$ (Ethyne / Acetylene) |
| Allylenides | $C_3^{4-}$ ($[:C=C=C:]^{4-}$) | $Mg_2C_3$ | $Mg_2C_3 + 4\ H_2O \rightarrow 2\ Mg(OH)_2 + \mathbf{CH3\text{-}C\equiv CH}\uparrow$ (Propyne / Allylene) |

Table 14.3 — Ionic carbides and their diagnostic hydrolysis gases (JEE Favourite).

KEY RESULT
Memory Rule for Hydrolysis:
- **$Be_2C$ & $Al_4C_3$** → CH4 (Methanides)
- **CaC2, BaC2, $Na_2C_2$** → $C_2H_2$ (Acetylides)
- **$Mg_2C_3$** → CH3-C≡CH (Allylenide / Propyne)

## 2. Covalent & Interstitial Carbides

- **Covalent Carbides:** Formed by elements of similar size and electronegativity:
  - **Silicon Carbide (SiC, 'Carborundum'):** 3D diamond-like covalent network. Extremely hard (Mohs 9.5), inert to acids, refractory abrasive (prep: SiO2 + 3 C —(2000°C)→ SiC + 2 CO).
  - **Boron Carbide ($B_4C$):** Hardest artificial carbide, used in tank armor and nuclear reactor control rods.
- **Interstitial Carbides:** Formed by transition metals (TiC, WC, TaC, $V_4C_3$, $Fe_3C$):
  Small carbon atoms occupy octahedral interstitial voids in the close-packed metal lattice without changing metal structure. Retain metallic electrical conductivity, possess ultra-high melting points (>3000°C), and extreme hardness.

**Oxides of Carbon** — CO (Ligand & Toxicity), CO2, $C_3O_2$ & Carbonate Chemistry

Carbon forms three well-characterised oxides: neutral CO, acidic CO2, and carbon suboxide $C_3O_2$.

## 1. Carbon Monoxide (CO) — Structure & Bonding

Iso-electronic with $N_2$ and $CN^-$ (10 valence electrons). Molecular Orbital configuration: $\sigma(2s)^2\ \sigma^*(2s)^2\ \pi(2p_x)^2\ \pi(2p_y)^2\ \sigma(2p_z)^2$. Bond order = 3 (one sigma, two pi bonds).

- **Dipole Moment:** Remarkably low (0.11 D) directed $C^{\delta-} \leftarrow O^{\delta+}$ (dative lone pair from O to C cancels electronegativity difference!).
- **Coordination Chemistry:** Acts as a strong **π-acid ligand** in transition metal carbonyls (e.g. $Ni(CO)_4$, $Fe(CO)_5$) via σ-donation from C and π-backbonding into empty π* orbitals of CO.
- **Toxicity:** Binds to blood haemoglobin ≈300 times more strongly than $O_2$ to form **carboxyhaemoglobin**, preventing oxygen delivery.

| Laboratory Synthesis | Industrial / Gas Mixtures | Reactions |
|---|---|---|
| Dehydration of Formic Acid: $HCOOH \xrightarrow{\text{(conc. } H_2SO_4,\ 373\ K)} CO + H_2O$ | Water Gas (Synthesis gas): $C + H_2O(g) \xrightarrow{(1270\ K)} CO + H_2$ | Combustion: $2\ CO + O_2 \rightarrow 2\ CO_2$ (blue flame) |
| Dehydration of Oxalic Acid: $(COOH)_2 \xrightarrow{\text{(conc. } H_2SO_4)} CO + CO_2 + H_2O$ (absorb CO2 in KOH) | Producer Gas: $2\ C + O_2 + 4\ N_2 \xrightarrow{(1270\ K)} 2\ CO + 4\ N_2$ | Mond Process: $Ni + 4\ CO \xrightarrow{(330\ K)} Ni(CO)_4 \xrightarrow{(450\ K)} Ni + 4\ CO$ |

Table 14.4 — Synthesis and reactions of Carbon Monoxide.

## 2. Carbon Suboxide (C3O2)

Prepared by dehydrating malonic acid with phosphorus pentoxide:
$CH_2(COOH)_2 + P_4O_{10} \xrightarrow{(150°C)} \mathbf{O{=}C{=}C{=}C{=}O} + 4\ HPO_3$
Structure: Linear, cumulative double bond system, sp-hybridized central carbons, bp 7°C, foul odor. Hydrolyses back to malonic acid.

## 3. Carbon Dioxide (CO2) & Bicarbonate Buffers

- Linear molecule (O=C=O, $D_{\infty h}$ symmetry), μ = 0, C-O bond length 116 pm (partial triple bond character).
- Lime water test: $Ca(OH)_2 + CO_2 \rightarrow \mathbf{CaCO_3}\downarrow$ (white turbidity) + $H_2O$. Excess CO2 dissolves turbidity: $CaCO_3 + CO_2 + H_2O \rightarrow \mathbf{Ca(HCO_3)_2}$ (soluble). On boiling, turbidity returns!

**Silicon Dioxide (SiO2) & Silanes** — 3D Network Structure & Hydrolysis

While CO2 is a discrete linear gas, SiO2 is a high-melting 3D network solid because Si cannot form stable pπ-pπ double bonds with oxygen.

## 1. Structure & Allotropy of Silica (SiO2)

- **Network Architecture:** Every Si atom is **$sp^3$ hybridized** and tetrahedrally bonded to 4 Oxygen atoms. Every O atom is shared between 2 Si atoms (Si-O-Si angle ≈ 144°), yielding empirical formula SiO2.
- **Crystalline Forms:** Quartz (piezoelectric), Tridymite, and Cristobalite (interconvertible at high temperatures).
- **Chemical Inertness:** Very inert; attacked only by HF and strong hot caustic alkali:
  · With HF (Etching of glass): $SiO_2 + 4\ HF \rightarrow$ **$SiF_4$** $+ 2\ H_2O$; $SiF_4 + 2\ HF \rightarrow$ **$H_2SiF_6$** (Hydrofluorosilicic acid).
  · With NaOH: $SiO_2 + 2\ NaOH \rightarrow$ **$Na_2SiO_3$** (Water glass) $+ H_2O$.

## 2. Silanes ($Si_nH_{2n+2}$) vs Alkanes

Silanes are the silicon hydrides ($SiH_4$, $Si_2H_6$, up to $Si_6H_{14}$), prepared by reacting $Mg_2Si$ with dil. HCl: $Mg_2Si + 4\ HCl \rightarrow 2\ MgCl_2 + SiH_4$ (Monosilane).

| Property | Methane ($CH_4$) | Silane ($SiH_4$) |
|---|---|---|
| Combustion / Air Stability | Stable in air, requires ignition ($\Delta H_{comb}$ = -890 kJ) | Spontaneously inflammable in air (pyrophoric!): $SiH_4 + 2\ O_2 \rightarrow SiO_2 + 2\ H_2O$ |
| Hydrolysis by Water | Completely inert to water and aqueous alkali | Violently hydrolysed by even traces of base: $SiH_4 + (n+2)\ H_2O \xrightarrow{(OH^-)} SiO_2{\cdot}nH_2O + 4\ H_2\uparrow$ |
| Polarity of E-H Bond | $C^{\delta-} - H^{\delta+}$ ($\chi_C$ = 2.5 > $\chi_H$ = 2.1) | $Si^{\delta+} - H^{\delta-}$ ($\chi_{Si}$ = 1.8 < $\chi_H$ = 2.1; Hydridic H!) |
| Mechanism of Attack | Carbon has no vacant d orbitals; SN2 hindered | Si has accessible empty 3d orbitals; nucleophilic attack by $OH^-$ is very fast |

Table 14.5 — Why Silane is hyper-reactive compared to Methane.

TRAP
TRAP: Carbon tetrachloride $CCl_4$ is completely inert to water, but Silicon tetrachloride $SiCl_4$ hydrolyses violently: $SiCl_4 + 4\ H_2O \rightarrow Si(OH)_4 + 4\ HCl$. Si utilizes vacant 3d orbitals to expand its coordination number!

**Silicate Minerals** — Classification, Structural Units & Sharing of Oxygen Vertices

The basic building unit of all silicates is the tetrahedral **$[SiO_4]^{4-}$** unit. Silicates are classified strictly by how many corners (oxygen atoms) are shared between adjacent $[SiO_4]$ tetrahedra.

## 1. Classification of Silicates

| Silicate Class | Shared O / Tetrahedron | General Formula Unit | Mineral Examples |
|---|---|---|---|
| 1. Orthosilicates (Neso) | 0 (discrete units) | $[SiO_4]^{4-}$ | Phenacite ($Be_2SiO_4$), Willemite ($Zn_2SiO_4$), Olivine ($Mg_2SiO_4$), Forsterite |
| 2. Pyrosilicates (Soro) | 1 (dimeric unit) | $[Si_2O_7]^{6-}$ | Thortveitite ($Sc_2Si_2O_7$), Hemimorphite ($Zn_4(OH)_2Si_2O_7·H_2O$) |
| 3. Cyclic / Ring | 2 (closed rings) | $[Si_3O_9]^{6-}$ or $[Si_6O_{18}]^{12-}$ | Benitoite ($BaTiSi_3O_9$), Beryl ($Be_3Al_2Si_6O_{18}$ — gemstone emerald) |
| 4. Single Chain (Ino / Pyroxenes) | 2 (infinite line) | $([SiO_3]^{2-})_n$ | Diopside ($CaMg(SiO_3)_2$), Spodumene ($LiAl(SiO_3)_2$), Enstatite |
| 5. Double Chain (Amphiboles) | 2.5 (alternating 2 and 3) | $([Si_4O_{11}]^{6-})_n$ | Tremolite ($Ca_2Mg_5(Si_4O_{11})_2(OH)_2$), Asbestos minerals |
| 6. Sheet Silicates (Phyllo) | 3 (2D planar infinite sheets) | $([Si_2O_5]^{2-})_n$ | Talc ($Mg_3(Si_2O_5)_2(OH)_2$), Kaolinite (clay), Mica, Muscovite |
| 7. 3D Framework (Tecto) | 4 (all four corners shared) | $(SiO_2)_n$ | Quartz, Feldspar, Zeolites, Ultramarines |

Table 14.6 — Master classification of Silicate minerals (JEE Advanced core).

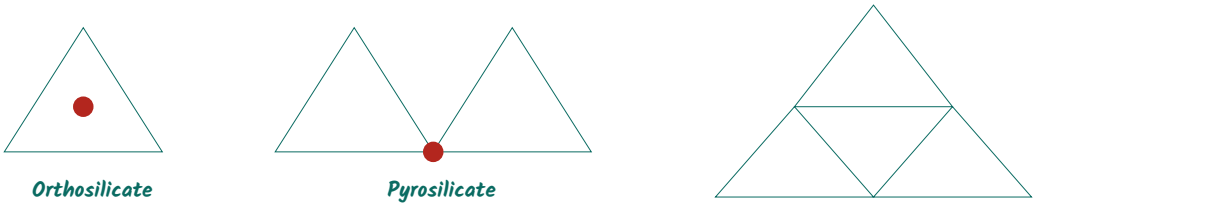



Fig. 14.1 — Structural polyhedra of Ortho, Pyro, and Cyclic silicates.

KEY RESULT

General Charge Rule: Formula of a silicate sharing 'c' corners per tetrahedron is **$[SiO_{4-c/2}]^{(4-c)-}$**.

- $c = 0 \rightarrow [SiO_4]^{4-}$ (Ortho)
- $c = 1 \rightarrow [Si_2O_7]^{6-}$ (Pyro)
- $c = 2 \rightarrow [SiO_3]_n^{2n-}$ (Cyclic & Single Chain)
- $c = 3 \rightarrow [Si_2O_5]_n^{2n-}$ (Sheet)
- $c = 4 \rightarrow SiO_2$ (3D Framework)

**Silicones & Zeolites** — Synthetic Polymers & Shape-Selective Catalysts

Silicones are synthetic organosilicon polymers containing repeating **$-R_2Si-O-$** units. Zeolites are microporous aluminosilicate 3D framework molecular sieves.

## 1. Synthesis & Control of Silicones

Prepared from alkylchlorosilanes synthesized by Rochow reaction: $2\ RCl + Si \xrightarrow{Cu\ powder,\ 570\ K} R_2SiCl_2$ :

- **Chain Propagation (Linear Silicones):** Hydrolysis of **$R_2SiCl_2$** gives $R_2Si(OH)_2$ which undergoes linear condensation polymerization:
  $n\ R_2SiCl_2 + 2n\ H_2O \rightarrow n\ R_2Si(OH)_2 \rightarrow$ **$...-O-Si(R_2)-O-Si(R_2)-O-...$** (linear silicone).
- **Chain Termination (Controlling Molecular Weight):** Addition of **$R_3SiCl$** produces monofunctional $R_3Si-OH$, capping the chain ends and terminating growth.
- **Cross-Linked (Resin) Silicones:** Hydrolysis of **RSiCl3** produces trifunctional RSi(OH)3 which condenses in three directions forming a rigid 3D cross-linked silicone polymer.

| Monomer Precursor | Hydrolysis Intermediate | Resulting Silicone Polymer Structure |
|---|---|---|
| $R_3SiCl$ (monochlorosilane) | $R_3SiOH$ (monofunctional) | Dimer: $R_3Si-O-SiR_3$ (Chain Stopper / Capping agent) |
| $R_2SiCl_2$ (dichlorosilane) | $R_2Si(OH)_2$ (difunctional) | Linear chain polymer or Cyclic oligomers (e.g. $(R_2SiO)_3$) |
| RSiCl3 (trichlorosilane) | RSi(OH)3 (trifunctional) | Cross-linked 3D network resin (high thermal rigidity) |

Table 14.7 — Role of chlorosilane precursors in silicone architecture.

## 2. Properties & Applications of Silicones

High thermal stability, water-repellent (hydrophobic organic R groups point outward), electrical insulators, biocompatible (prosthetics, lubricants, waterproof fabrics).

## 3. Zeolites: Three-Dimensional Aluminosilicates

- **Formula & Charge Balance:** In 3D framework SiO2, when some $Si^{4+}$ atoms are replaced by $Al^{3+}$, the framework acquires a net negative charge balanced by extra-framework cations ($Na^+$, $K^+$, $Ca^{2+}$): **$M_{x/n}[(AlO2)_x(SiO2)_y]\cdot z\ H_2O$**.
- **Shape-Selective Catalysis:** Zeolites possess open honeycomb cages with uniform pore sizes (3 to 10 Å). Only molecules of fitting dimensions can enter and react.
- **ZSM-5 (Zeolite Socony Mobil-5):** Converts alcohols directly into gasoline (hydrocarbons) by shape-selective dehydration: **$x\ CH3OH \xrightarrow{ZSM\text{-}5,\ 670\ K} Gasoline\ (C_n H_{2n+2}) + x\ H_2O$**.
- **Water Softening:** Permutit (hydrated sodium aluminosilicate $Na_2Al_2Si_2O_8 \cdot xH_2O$) exchanges $Na^+$ for $Ca^{2+}$ and $Mg^{2+}$ in hard water.

**Group 14 Halides & Redox Trends** — Tetrahalides vs Dihalides

Tetrahalides $MX_4$ ($sp^3$ tetrahedral) dominate for C, Si, Ge, and Sn, whereas dihalides $MX_2$ (bent/polymeric) become progressively stable descending to Pb due to the inert pair effect.

## 1. Tetrahalides: Hydrolysis Mechanisms & Exceptions

- **$CCl_4$ Inertness:** $CCl_4$ does not hydrolyse in neutral or basic water because Carbon is in the 2nd period: no vacant d orbitals are available to expand its octet to accept the nucleophilic lone pair of water.
- **$SiCl_4$, $GeCl_4$, $SnCl_4$ Hydrolysis:** Rapid and exothermic. Water coordinates to vacant d orbitals of the central atom forming a 5-coordinate intermediate, followed by elimination of HCl:
$SiCl_4 + 4\ H_2O \rightarrow [Si(OH)_4]$ (orthosilicic acid) $\rightarrow SiO2{\cdot}xH2O + 4\ HCl$.
- **Lead Tetrahalides:** $PbCl_4$ is a yellow oily liquid that readily decomposes to $PbCl_2 + Cl_2$. **$PbBr_4$ and $PbI_4$ do not exist** because $Pb^{4+}$ oxidises $Br^-$ and $I^-$ spontaneously.

| Halide | Physical State | Oxidation State | Key Chemical Reaction / Behaviour |
|---|---|---|---|
| $SnCl_2$ | White solid | +2 (reducing) | $SnCl_2 + 2\ FeCl_3 \rightarrow SnCl_4 + 2\ FeCl_2$; reduces $HgCl_2 \rightarrow Hg_2Cl_2\downarrow \rightarrow Hg\downarrow$ |
| $SnCl_4$ | Colorless liquid (fuming) | +4 | Forms hexachlorostannate with HCl: $SnCl_4 + 2\ HCl \rightarrow H_2[SnCl_6]$ |
| $PbCl_2$ | White crystalline solid | +2 (stable) | Insoluble in cold water, soluble in hot water (recrystallises as needles) |
| $PbCl_4$ | Yellow liquid (unstable) | +4 (oxidising) | Decomposes explosively on heating: $PbCl_4 \xrightarrow{(\Delta)} PbCl_2 + Cl_2\uparrow$ |

Table 14.8 — Comparative chemistry of Tin and Lead chlorides.

## 2. Dihalides & Carbene Analogues

In the gas phase, Group 14 dihalides ($CF_2$, $SiCl_2$, $GeCl_2$, $SnCl_2$, $PbCl_2$) are VSEPR **bent (V-shaped)** molecules with a lone pair and an empty p orbital (carbene analogues). In the solid state, $SnCl_2$ and $PbCl_2$ form extended halogen-bridged polymers or ionic lattices.

KEY RESULT

Oxidising vs Reducing Summary:

- **$Sn^{2+}$ is REDUCING:** $Sn^{2+} \rightarrow Sn^{4+} + 2e^-$ (favors +4 state)
- **$Pb^{4+}$ is OXIDISING:** $Pb^{4+} + 2e^- \rightarrow Pb^{2+}$ (favors +2 state due to inert pair effect)

**Group 15: Nitrogen Family ($ns^2\ np^3$)** — Pnictogens & Periodic Trends

Contains N, P (non-metals), As, Sb (metalloids), Bi (post-transition metal). The valence shell has a stable half-filled $p^3$ configuration, resulting in unusually high ionization energies.

## 1. Periodic Trends of Group 15

| Element | Config. | Covalent Radius (pm) | IE-1 (kJ/mol) | Electronegativity | Allotropes |
|---|---|---|---|---|---|
| N | [He] $2s^2\ 2p^3$ | 70 | 1402 | 3.0 | Dinitrogen gas (α, β solid forms) |
| P | [Ne] $3s^2\ 3p^3$ | 110 | 1012 | 2.1 | White, Red, Black (α, β), Violet |
| As | [Ar] $3d^{10}\ 4s^2\ 4p^3$ | 121 | 947 | 2.0 | Grey (metallic), Yellow, Black |
| Sb | [Kr] $4d^{10}\ 5s^2\ 5p^3$ | 141 | 834 | 1.9 | Metallic (silver-grey), Yellow, Black |
| Bi | [Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^3$ | 148 | 703 | 1.9 | White with pinkish tinge (only 1 form) |

Table 15.1 — Physical and atomic properties of Group 15 elements.

KEY RESULT
IE-1 is substantially higher than Group 16 elements (e.g., $IE_1(N) = 1402 > IE_1(O) = 1314$ kJ/mol) due to the extra stability of the half-filled $2p^3$ subshell in Nitrogen!

## 2. Anomalous Behaviour of Nitrogen

- **Small size & High Electronegativity:** Forms very strong **pπ-pπ triple bonds** (:N≡N: bond energy = 945.8 kJ/mol). Nitrogen exists as a chemically inert diatomic gas at RTP.
- **Single Bond Anomaly (N-N vs P-P):** N-N single bond energy (160 kJ/mol) is **much WEAKER** than P-P single bond energy (213 kJ/mol) due to intense inter-electronic repulsion between lone pairs in the compact 2p shell. Hence, **catenation in N is very limited** compared to P.
- **Absence of d-orbitals:** Nitrogen has a **maximum covalency of 4** (one 2s and three 2p orbitals). It CANNOT form pentahalides (NF5 does not exist) or expand octet like P ($PCl_5$, $PF_6^-$).

## 3. Oxidation States & Inert Pair Effect

Common states: -3, +3, +5. Moving down the group:
· **-3 state stability decreases:** $Bi^{3-}$ is virtually unknown due to metallic electropositive nature.
· **+5 state stability decreases:** Bi(+5) is unstable and a **powerful oxidising agent** ($BiF_5$ is known, but $BiCl_5$ does not exist; $NaBiO_3$ is a classic analytical oxidant for $Mn^{2+} \rightarrow MnO_4^-$).
· **+3 state stability increases:** Bi(+3) is the most stable state of Bismuth due to the **inert pair effect**.

**Dinitrogen ($N_2$)** — Preparation, Inertness, Fixation & Chemical Reactions

Constitutes 78% of air by volume. Colorless, odorless, diamagnetic gas, condensation temp 77.2 K.

## 1. Laboratory & High-Purity Preparations

| Method | Reaction Equation | Purity & Key Notes |
|---|---|---|
| Aqueous Nitrite Reaction | $NH_4Cl(aq) + NaNO_2(aq) \rightarrow \mathbf{N_2}\uparrow + NaCl + 2\ H_2O$ | Standard lab prep; contains trace NO and $HNO_3$ impurities (purified by passing through aq. $H_2SO_4 + K_2Cr_2O_7$) |
| Thermal Decomposition of Dichromate | $(NH_4)_2Cr_2O_7 \xrightarrow{(\Delta)} \mathbf{N_2}\uparrow + Cr_2O_3$ (green solid) $+ 4\ H_2O$ | Volcano reaction; exothermic decomposition of orange crystals |
| Thermal Azide Decomposition | $Ba(N_3)_2 \xrightarrow{(300°C)} Ba + \mathbf{3\ N_2}\uparrow$ or $2\ NaN_3 \rightarrow 2\ Na + \mathbf{3\ N_2}\uparrow$ | **Ultra-pure $N_2$ (>99.99%)**; sodium azide used in automobile airbags! |
| Oxidation of Ammonia | $8\ NH_3 + 3\ Cl_2$ (excess $NH_3$) $\rightarrow \mathbf{N_2}\uparrow + 6\ NH_4Cl$ or $2\ NH_3 + 3\ CuO \rightarrow \mathbf{N_2}\uparrow + 3\ Cu + 3\ H_2O$ | Yellow liquid $NCl_3$ formed if $Cl_2$ is in excess! |

Table 15.2 — Synthetic routes to Dinitrogen.

KEY RESULT
Ultra-Pure Nitrogen is prepared exclusively by the **thermal decomposition of sodium or barium azide** $NaN_3$ / $Ba(N_3)_2$.

## 2. Chemical Reactivity of N2

Remarkably unreactive at room temperature because of the immense bond dissociation enthalpy of N≡N (945.8 kJ/mol). Becomes active only at elevated temperatures:

- **With Metals (Nitrides):** $6\ Li + N_2 \rightarrow 2\ Li_3N$ (ruby red ionic solid); $3\ Mg + N_2 \rightarrow Mg_3N_2$ (reacts with water to evolve $NH_3$: $Mg_3N_2 + 6\ H_2O \rightarrow 3\ Mg(OH)_2 + 2\ NH_3\uparrow$).
- **With Non-Metals:**
  - $N_2 + 3\ H_2 \rightleftharpoons 2\ NH_3$ (Haber process, 700 K, 200 atm, Fe/Mo catalyst, $\Delta H = -92.4$ kJ/mol).
  - $N_2 + O_2 \rightleftharpoons 2\ NO$ (Electric arc, T > 2000 K, Birkeland-Eyde process, endothermic $\Delta H = +180$ kJ/mol).
- **Nitrolim Formation:** $CaC_2 + N_2 \xrightarrow{(1373\ K)} \mathbf{CaCN_2 + C}$ ('Nitrolim' or Calcium Cyanamide). Used as an agricultural fertilizer; slowly hydrolyses in soil to urea and ammonia: $CaCN_2 + 3\ H_2O \rightarrow CaCO_3 + 2\ NH_3$.

**Ammonia (NH3)** — Haber Synthesis, Molecular Properties & Liquid NH3 Solvent

Trigonal pyramidal geometry ($sp^3$ hybridized, one lone pair, bond angle 107.8°, dipole moment 1.47 D). Associated in liquid and solid states via intermolecular hydrogen bonding.

## 1. Synthesis: The Haber-Bosch Process

KEY RESULT
$N_2(g) + 3\ H_2(g) \rightleftharpoons 2\ NH3(g)$ $\Delta H° = -92.4$ kJ/mol (Exothermic)
Optimum Le Chatelier conditions: High pressure (≈200 atm), Moderate temperature (≈700 K), Catalyst: Iron oxide ($FeO/Fe_2O_3$) with promoter $K_2O$ and $Al_2O_3$ (increases surface area and activity).

## 2. Chemical Reactions of NH3

- **Basic Nature & Salt Formation:** Weak base in water ($K_b = 1.8 \times 10^{-5}$^). Forms ammonium salts with acids (NH3 + HCl → NH4Cl white dense fumes).
- **Precipitation of Metal Hydroxides:** $Fe^{3+} + 3\ NH4OH \rightarrow Fe(OH)_3\downarrow$ (reddish brown); $Zn^{2+} + 2\ NH4OH \rightarrow Zn(OH)_2\downarrow$ (white, dissolves in excess NH3 forming $[Zn(NH_3)_4]^{2+}$).
- **Complexation with Transition Metals (Deep Blue / Colour Changes):**
  · $Cu^{2+}(aq)$ (pale blue) + 4 NH3 → **$[Cu(NH_3)_4]^{2+}$** (tetraamminecopper(II), deep Prussian blue).
  · $AgCl\downarrow$ (white precipitate) + 2 NH3 → **$[Ag(NH_3)_2]^+ + Cl^-$** (soluble diamminesilver(I)).
- **Nessler's Reagent Test:** Alkaline $K_2[HgI4]$ + NH3 → **$[HgO{\cdot}Hg(NH_2)I]\downarrow$** (Iodide of Millon's base, brown precipitate). Extremely sensitive qualitative test for $NH4^+$.

## 3. Liquid Ammonia as a Non-Aqueous Ionising Solvent

Liquid NH3 (bp -33°C) autoionizes: $2\ NH3 \rightleftharpoons NH4^+$ (acid analogue) + $NH2^-$ (base analogue). Alkali metals dissolve in liquid ammonia giving spectacular solutions:

| Concentration | Colour & Magnetic Property | Active Species & Behaviour |
|---|---|---|
| Dilute Solution (<3 M) | **Deep Blue**, strongly **paramagnetic** | **Ammoniated electrons $[e(NH3)_y]^-$**: Powerful reducing agent (Birch reduction) |
| Concentrated (>3 M) | **Bronze / Copper luster**, **diamagnetic** | Electrons pair up to form diamagnetic electron pairs; metallic electrical conductivity |

Table 15.3 — Metal-ammonia solutions (NCERT & JD Lee core).

**Nitrogen Hydrides & Derivatives** — $N_2H_4$, NH2OH, HN3 & Redox Chemistry

Nitrogen forms a remarkable sequence of hydrides spanning oxidation states from -3 to -1/3.

## 1. Hydrazine (N2H4, H2N-NH2, Oxidation State = -2)

- **Raschig Synthesis:** 2 NH3 + NaOCl —(gelatin/glue catalyst)→ **$N_2H_4$** + NaCl + $H_2O$.
  (Gelatin complexes with heavy metal ions like $Cu^{2+}$ which would otherwise catalyse hydrazine decomposition!).
- **Structure:** Gauche conformation (dihedral angle ≈ 90° to minimise lone-pair repulsions, $sp^3$ N atoms).
- **Reactions & Uses:** Fuming colorless liquid, bifunctional weak base ($pK_{b1}$ = 6.07). Powerful **reducing agent**: $N_2H_4$ + 2 $I_2$ → $N_2$↑ + 4 HI (quantitative, no residue; hence used as rocket propellant fuel with $N_2O_4$ ).

## 2. Hydroxylamine (NH2OH, Oxidation State = -1)

Prepared by electrolytic reduction of nitric acid: HNO3 + 6 $H^+$ + 6 $e^-$ → **NH2OH** + 2 $H_2O$.
Acts as both oxidising and reducing agent:
· Reducing action: 2 NH2OH + 4 $Fe^{3+}$ → $N_2O$ + 4 $Fe^{2+}$ + 4 $H^+$ + $H_2O$.
· Oxidising action: NH2OH + 2 $Fe^{2+}$ + 2 $H^+$ → NH3 + 2 $Fe^{3+}$ + $H_2O$.
Forms **oximes** with aldehydes and ketones: $R_2C=O$ + NH2OH → $R_2C=N-OH$ + $H_2O$.

## 3. Hydrazoic Acid & Azides (HN3, Oxidation State = -1/3)

| Compound / Ion | Formal Oxidation State | Structure & Geometry | Safety / Stability |
|---|---|---|---|
| Hydrazoic acid HN3 | -1/3 | $H-N_1=N_2=N_3$ (linear N-N-N backbone, angle 171°) | Colorless, volatile liquid (bp 37°C), violently explosive! |
| Azide ion $N_3^-$ | -1/3 | $[:N=N=N:]^-$ (symmetric linear, $D_{\infty h}$, N-N bond 116 pm) | $Pb(N_3)_2$ used in percussion caps and explosive detonators |

Table 15.4 — Azide systems in inorganic chemistry.

KEY RESULT
Oxidation State Summary of Nitrogen Hydrides:
· **NH3**: -3 | **$N_2H_4$**: -2 | **NH2OH**: -1 | **$N_2$**: 0 | **HN3**: -1/3

**Oxides of Nitrogen** — Structures, Oxidation States, Colors & Magnetic Properties

Nitrogen forms six classic oxides across oxidation states from +1 to +5. Several possess odd electrons.

## 1. Comprehensive Summary of Nitrogen Oxides

| Oxide | Oxid. State | Prep Reaction | Colour & Physical State | Magnetic Nature |
|---|---|---|---|---|
| $N_2O$ (Nitrous oxide) | +1 | $NH_4NO_3 \xrightarrow{(\Delta,\ 250°C)} N_2O + 2\ H_2O$ | Colorless gas (Laughing gas) | Diamagnetic (Neutral) |
| NO (Nitric oxide) | +2 | $2\ NaNO_2 + 2\ FeSO_4 + 3\ H_2SO_4 \rightarrow Fe_2(SO_4)_3 + 2\ NaHSO_4 + 2\ H_2O + 2\ NO$ | Colorless gas (Liquid is blue) | **Paramagnetic (gas)**, Diamag (liquid/solid dimer) (Neutral) |
| $N_2O_3$ (Sesquioxide) | +3 | $2\ NO + N_2O_4 \xrightarrow{(250\ K)} 2\ N_2O_3$ | **Deep Blue liquid** / solid | Diamagnetic (Acidic) |
| $NO_2$ (Dioxide) | +4 | $2\ Pb(NO_3)_2 \xrightarrow{(\Delta,\ 673\ K)} 2\ PbO + 4\ NO_2 + O_2$ | **Brown gas** | **Paramagnetic** (odd-electron molecule) (Acidic) |
| $N_2O_4$ (Tetroxide) | +4 | $2\ NO_2 \xleftrightarrow{(cool\ /\ warm)} N_2O_4$ | Colorless liquid / solid | Diamagnetic (Acidic) |
| $N_2O_5$ (Pentoxide) | +5 | $4\ HNO_3 + P_4O_{10} \rightarrow 4\ HPO_3 + 2\ N_2O_5$ | Colorless crystalline solid | Diamagnetic (Strongly Acidic) |

Table IS.5 — Complete spectrum of Nitrogen Oxides (NCERT & JD Lee core).

KEY RESULT
Neutral Oxides: **$N_2O$ and NO** are completely NEUTRAL to litmus! All other oxides ($N_2O_3$, $NO_2$, $N_2O_4$, $N_2O_5$) are acidic anhydrides.

## 2. Bonding & Structural Details

- **NO (Paramagnetic Monomer vs Diamagnetic Dimer):** Gas contains 11 valence electrons; unpaired electron resides in a π antibonding molecular orbital (bond order 2.5). In liquid and solid states, it dimerises to planar cis-O=N-N=O, pairing electrons (diamagnetic).
- **$NO_2 \rightleftharpoons N_2O_4$ Equilibrium:** $NO_2$ has a single odd electron on N (brown, paramagnetic, bent O-N-O angle 134°). On cooling, two $NO_2$ molecules dimerise to form planar $N_2O_4$ with an unusually long N-N bond (175 pm vs normal 145 pm) due to repulsion between bonding pairs.
- **Solid $N_2O_5$ Ionic Lattice:** In the gas phase, $N_2O_5$ is molecular ($O_2N$-O-$NO_2$). In the **solid crystalline state**, it exists as an ionic salt: **$[NO_2]^+$ $[NO_3]^-$ (Nitronium nitrate)**! Linear $[NO_2]^+$ cation + planar trigonal $[NO_3]^-$ anion.

TRAP
TRAP: Liquid $N_2O_4$ undergoes auto-ionization: $N_2O_4 \rightleftharpoons NO^+$ (nitrosyl cation) + $NO_3^-$ (nitrate anion). It acts as a non-aqueous solvent where NOCl acts as an acid and $AgNO_3$ acts as a base!

**Oxoacids of Nitrogen** — Nitrous Acid, Nitric Acid & The Brown Ring Test Mechanism

Primary oxoacids: Nitrous acid HNO2 (+3, unstable, weak acid) and Nitric acid HNO3 (+5, strong acid & oxidising agent).

## 1. Nitrous Acid (HNO2) & Disproportionation

Prepared freshly in cold: $NaNO_2 + HCl \xrightarrow{(0\text{-}5^\circ C)} NaCl + HNO_2$. On warming, it **disproportionates**:
$3\ HNO_2 \xrightarrow{(warm)} \mathbf{HNO_3\ (+5) + 2\ NO\ (+2) + H_2O}$.
Acts as both oxidising agent (oxidises $I^-$ to $I_2$) and reducing agent (decolourises $KMnO_4$).

## 2. Nitric Acid (HNO3) Manufacture: Ostwald Process

- **Step 1 (Catalytic Oxidation):** $4\ NH_3(g) + 5\ O_2(g) \xrightarrow{(Pt\text{-}Rh\ gauze,\ 1150\ K,\ 9\ bar)} \mathbf{4\ NO(g)} + 6\ H_2O(g)$ (exothermic).
- **Step 2 (Dimerization / Oxidation):** $2\ NO(g) + O_2(g) \rightleftharpoons \mathbf{2\ NO_2(g)}$.
- **Step 3 (Absorption in Water):** $3\ NO_2(g) + H_2O(l) \rightarrow \mathbf{2\ HNO_3(aq) + NO(g)}$ (NO is recycled back to Step 2).

## 3. Oxidising Action of HNO3 on Metals

| Metal | Concentration of HNO3 | Oxidation Product & Nitrogen Gas Evolved |
|---|---|---|
| Copper (Cu) | Dilute HNO3 (≈20%) | $3\ Cu + 8\ HNO_3 \rightarrow 3\ Cu(NO_3)_2 + 2\ H_2O + \mathbf{2\ NO}\uparrow$ (Nitric oxide) |
| Copper (Cu) | Concentrated HNO3 (≈70%) | $Cu + 4\ HNO_3 \rightarrow Cu(NO_3)_2 + 2\ H_2O + \mathbf{2\ NO_2}\uparrow$ (Brown gas) |
| Zinc (Zn) | Dilute HNO3 (≈20%) | $4\ Zn + 10\ HNO_3 \rightarrow 4\ Zn(NO_3)_2 + 5\ H_2O + \mathbf{N_2O}\uparrow$ (Nitrous oxide) |
| Zinc (Zn) | Concentrated HNO3 (≈70%) | $Zn + 4\ HNO_3 \rightarrow Zn(NO_3)_2 + 2\ H_2O + \mathbf{2\ NO_2}\uparrow$ (Brown gas) |
| Zinc (Zn) | Very Dilute HNO3 (≈6%) | $4\ Zn + 10\ HNO_3 \rightarrow 4\ Zn(NO_3)_2 + 3\ H_2O + \mathbf{NH_4NO_3}$ (Ammonium nitrate) |

Table 15.6 — Reaction of metals with dilute vs concentrated Nitric Acid (JEE Classic).

## 4. Mechanism of the Brown Ring Test for Nitrate ($NO_3^-$)

Aqueous nitrate + freshly prepared $FeSO_4$ + conc. $H_2SO_4$ carefully poured down the side of the test tube:
1. Reduction: $NO_3^- + 3\ Fe^{2+} + 4\ H^+ \rightarrow 3\ Fe^{3+} + \mathbf{NO} + 2\ H_2O$.
2. Coordination & Complex Formation: $[Fe(H_2O)_6]^{2+} + NO \rightarrow \mathbf{[Fe(H_2O)_5(NO)]^{2+}}$ (Brown Ring) $+ H_2O$.

KEY RESULT
**Oxidation State of Fe in Brown Ring Complex $[Fe(H_2O)_5(NO)]^{2+}$:**
One electron is transferred from NO to $Fe^{2+}$, creating **Fe(+1) ($d^7$, $t2g^5\ eg^2$)** and **$NO^+$ (nitrosonium ion)**!
Magnetic Moment: $\mu = \sqrt{15} \approx$ **3.87 BM (3 unpaired electrons)**. This is a critical JEE Advanced question.

**Phosphorus Allotropes & Phosphine** — White, Red, Black Allotropy & PH3 Smoke Rings

Phosphorus exists in three major allotropic forms: White ($P_4$), Red (polymeric), and Black (layered).

## 1. Comparative Allotropy of Phosphorus

| Property | White Phosphorus | Red Phosphorus | Black Phosphorus (α and β) |
|---|---|---|---|
| Molecular Structure | Discrete $P_4$ tetrahedra (angle 60°) | Complex 3D polymer of linked $P_4$ rings | Layered puckered hexagonal sheets (flaky) |
| Physical State | Translucent waxy solid (soft, cuts with knife) | Reddish-violet powdery solid (odorless) | Black lustrous flaky solid (graphite-like) |
| Toxicity & Chemiluminescence | **Highly toxic**, glows green in dark (chemiluminescence) | **Non-toxic**, does NOT glow in dark | Non-toxic, completely non-reactive |
| Reactivity & Ignition | Spontaneously catches fire in air at 30°C (kept under water!) | Stable in air, ignites at 260°C | Least reactive, highly inert |
| Solubility | Insoluble in $H_2O$, soluble in CS2 and benzene | Insoluble in both $H_2O$ and CS2 | Insoluble in all ordinary solvents |

Table 15.7 — Comparison of Phosphorus allotropes (NCERT core).

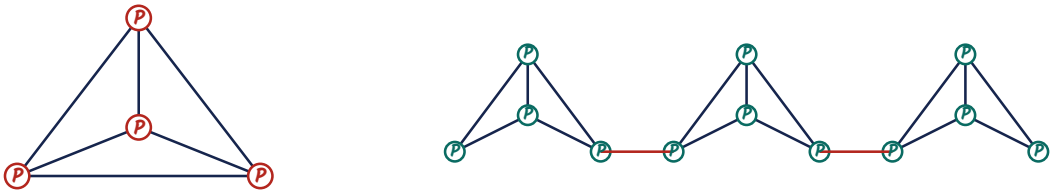



Fig. 15.1 — White $P_4$ discrete tetrahedron vs Red Phosphorus polymeric chain.

## 2. Phosphine ($PH_3$) — Synthesis & Reactions

- **Preparation (Disproportionation of White $P_4$):**
  $P_4 + 3\ NaOH + 3\ H_2O$ —(inert CO2 atm)→ **PH3↑ + 3 NaH2PO2** (Sodium hypophosphite, +1 state).
  (Pure PH3 is non-flammable; it catches fire spontaneously due to traces of **$P_2H_4$ (diphosphane)** impurity!).
- **Holmes Smoke Signals:** Containers of CaC2 + $Ca_3P_2$ pierced and thrown into sea. Water reacts:
  $Ca_3P_2 + 6\ H_2O \rightarrow 3\ Ca(OH)_2 + 2\ PH_3\uparrow$ (contains $P_2H_4$ → ignites → produces vortex smoke rings of $P_4O_{10}$).
  $CaC_2 + 2\ H_2O \rightarrow Ca(OH)_2 + C_2H_2\uparrow$ (acetylene burns giving bright illuminating flame).
- **Basicity:** PH3 is a very weak base (much weaker than NH3, no H-bonding, Drago's rule: unhybridized pure p-orbital lone pair). Only forms salts with strong acids: PH3 + HI → **PH4I** (Phosphonium iodide).

**Phosphorus Halides — PCl3, PCl5, Axial/Equatorial Bonds & Solid-State Ionization**

Phosphorus forms trihalides PX3 ($sp^3$ pyramidal) and pentahalides PX5 ($sp^3d$ trigonal bipyramidal).

## 1. Phosphorus Trichloride (PCl3)

- **Synthesis:** $P_4$ + 6 $Cl_2$ (dry, limited) → 4 PCl3 (or: $P_4$ + 8 SOCl2 → 4 PCl3 + 4 SO2 + 2 $S_2Cl_2$).
- **Structure:** Pyramidal ($sp^3$ P, one lone pair, P-Cl bond length 204 pm).
- **Hydrolysis:** PCl3 + 3 $H_2O$ → **$H_3PO_3$** + 3 HCl (violent, fumes copiously in moist air).
- **Reactions with Organic OH:** 3 $C_2H_5OH$ + PCl3 → 3 $C_2H_5Cl$ + $H_3PO_3$; 3 CH3COOH + PCl3 → 3 CH3COCl + $H_3PO_3$.

## 2. Phosphorus Pentachloride (PCl5) — Gas vs Solid State

| State of Matter | Chemical Structure & Species Present | Bond Lengths & Geometry |
|---|---|---|
| Gas Phase & Liquid | Discrete molecular PCl5 | Trigonal bipyramidal ($sp^3d$): **3 equatorial bonds (202 pm) + 2 axial bonds (240 pm)**. Axial bonds are longer and weaker due to 3 × 90° repulsions! |
| Solid Crystalline State | **Ionic Solid:** $[PCl_4]^+$ $[PCl_6]^-$ | $[PCl_4]^+$ is **tetrahedral ($sp^3$)**; $[PCl_6]^-$ is **octahedral ($sp^3d^2$)**! |
| PBr5 Solid State | **Ionic Solid:** $[PBr_4]^+$ $[Br]^-$ | $[PBr_4]^+$ tetrahedral cation + $Br^-$ anion (P is too small to coordinate 6 bulky Br atoms!) |

Table 15.8 — Gas-phase vs Solid-state structures of Phosphorus pentahalides.

KEY RESULT

Why are axial bonds longer than equatorial bonds in PCl5?

Equatorial P-Cl bonds experience repulsion from only two adjacent bonds at 90°. Each axial P-Cl bond experiences repulsion from **three equatorial bonds at 90°**, making them longer and weaker!

## 3. Hydrolysis of PCl5

Stepwise hydrolysis:

· Step 1 (Limited water): PCl5 + $H_2O$ → **POCl3 (Phosphorus oxychloride)** + 2 HCl.

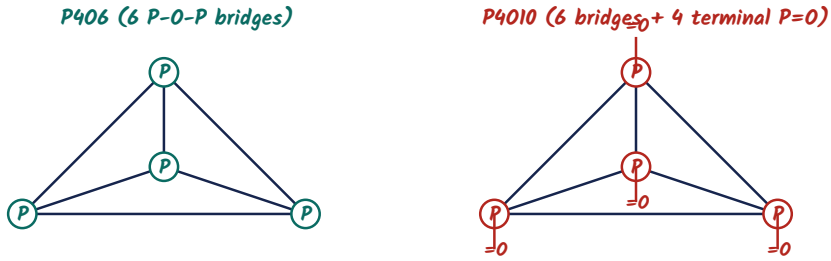
· Step 2 (Excess water): POCl3 + 3 $H_2O$ → **$H_3PO_4$ (Orthophosphoric acid)** + 3 HCl.

Net: **PCl5 + 4 $H_2O$ → $H_3PO_4$ + 5 HCl**.

**Oxides of Phosphorus** — $P_4O_6$, $P_4O_{10}$, Adamantane Cages & Dehydrating Power

Prepared by burning white phosphorus in air/oxygen: limited air gives $P_4O_6$, excess air gives $P_4O_{10}$.

## 1. Structure & Bonding of P4O6 and P4O10




**Each P in P4O10 is sp3 tetrahedral; 16 total P-O bonds (6 bridging P-O-P + 4 P=O)**

Fig. 15.2 — Adamantane cage architecture of $P_4O_6$ and $P_4O_{10}$.

| Feature | Phosphorus Trioxide ($P_4O_6$) | Phosphorus Pentoxide ($P_4O_{10}$) |
|---|---|---|
| Combustion Condition | $P_4 + 3\ O_2$ (limited) → $P_4O_6$ | $P_4 + 5\ O_2$ (excess) → $P_4O_{10}$ |
| Formal Oxidation State | +3 | +5 |
| Number of P-O-P Bridges | 6 bridging P-O-P bonds | 6 bridging P-O-P bonds |
| Terminal P=O Bonds | 0 (each P retains one lone pair) | 4 terminal P=O bonds (dπ-pπ backbonding) |
| Total Number of P-O Bonds | 12 P-O single bonds | **16 P-O bonds** (12 bridging + 4 terminal) |
| Hybridization of P | $sp^3$ (pyramidal at each P) | $sp^3$ (tetrahedral at each P) |

Table 15.9 — Structural parameters of $P_4O_6$ vs $P_4O_{10}$ cages.

KEY RESULT

Counting Bonds in $P_4O_{10}$ (Guaranteed JEE Question):

- Bridging P-O-P linkages = **6**
- Terminal P=O double bonds = **4**
- Total P-O single bond vectors in cage = **12** (plus 4 terminal = 16 total)

## 2. Hydrolysis Pathways & Acid Anhydrides

- **Hydrolysis of $P_4O_6$:** $P_4O_6 + 6\ H_2O$ (cold) → **$4\ H_3PO_3$ (Phosphorous acid)**.
  (With hot water, disproportionates: $P_4O_6 + 6\ H_2O \rightarrow 3\ H_3PO_4 + PH3\uparrow$).
- **Hydrolysis of $P_4O_{10}$:** Stepwise hydration:
  $P_4O_{10}$ —(+2 $H_2O$)→ 4 HPO3 (Metaphosphoric acid) —(+2 $H_2O$)→ 2 $H_4P_2O_7$ (Pyrophosphoric acid) —(+2 $H_2O$)→ **$4\ H_3PO_4$ (Orthophosphoric acid)**.
- **Powerful Dehydrating Action of $P_4O_{10}$:** It strips water from acids to generate anhydrides:
  - 2 HNO3 + $P_4O_{10}$ → **$N_2O_5$** + 2 HPO3
  - 2 $H_2SO_4$ + $P_4O_{10}$ → **2 SO3** + 4 HPO3
  - 2 CH3CONH2 + $P_4O_{10}$ → **2 CH3CN (Acetonitrile)** + 4 HPO3
  - (COOH)2 + $P_4O_{10}$ → **CO + CO2** + 2 HPO3

**Oxoacids of Phosphorus** — Basicity, P-H Bonds & Reducing Power

Phosphorus forms a famous series of oxoacids characterized by tetrahedral $sp^3$ P atoms with at least one P=O bond and one P-OH bond. Basicity equals the number of P-OH bonds!

## 1. Comprehensive Oxoacids Table

| Oxoacid | Formula | State | Basicity | No. of P-H | Reducing Character |
|---|---|---|---|---|---|
| Hypophosphorous acid | $H_3PO_2$ | +1 | **1 (Monobasic)** | **2 P-H bonds** | Very Strong (reduces AgNO3 → Ag↓) |
| Orthophosphorous acid | $H_3PO_3$ | +3 | **2 (Dibasic)** | **1 P-H bond** | Strong (reduces CuSO4 → Cu↓) |
| Orthophosphoric acid | $H_3PO_4$ | +5 | **3 (Tribasic)** | **0 P-H bonds** | Non-reducing |
| Pyrophosphoric acid | $H_4P_2O_7$ | +5 | **4 (Tetrabasic)** | 0 P-H bonds | Non-reducing (P-O-P bridge) |
| Hypophosphoric acid | $H_4P_2O_6$ | +4 | **4 (Tetrabasic)** | 0 P-H bonds | Non-reducing (**P-P direct bond!**) |
| Cyclotrimetaphosphoric | (HPO3)3 | +5 | **3 (Tribasic)** | 0 P-H bonds | Non-reducing (3 P-O-P bridges) |

Table 15.10 — Structural formulas and properties of Phosphorus oxoacids.

KEY RESULT

Basicity vs Reducing Rule:

- **Basicity = Number of P-OH bonds** (only H attached to oxygen is ionisable as $H^+$).
- **Reducing Power = Number of P-H bonds** (hydrogenic H attached directly to P is reducing!).

## 2. Diagnostic Disproportionation of Intermediate Acids

Acids containing phosphorus in intermediate oxidation states (+1 to +4) **disproportionate** on heating:

- 4 $H_3PO_3$ (+3) —(Δ, 200°C)→ **3 $H_3PO_4$ (+5) + $PH_3$ (-3)**
- 2 $H_3PO_2$ (+1) —(Δ, 160°C)→ **$H_3PO_4$ (+5) + $PH_3$ (-3)**

## 3. Quantitative Reaction with Silver Nitrate

Hypophosphorous acid $H_3PO_2$ has two P-H bonds and acts as a powerful analytical reducing agent:

4 AgNO3 + 2 $H_2O$ + $H_3PO_2$ → **4 Ag↓ (black metallic mirror) + $H_3PO_4$ + 4 HNO3**.

**Group 16: Chalcogens ($ns^2\ np^4$)** — Ore Formers & Periodic Trends

Contains O, S (non-metals), Se, Te (metalloids), Po (radioactive α-emitter metal). The electronic configuration has two unpaired p-electrons. Key themes: anomalous oxygen properties, catenation in sulphur, and pπ-dπ vs pπ-pπ multiple bonding.

## 1. Periodic Trends of Group 16

| Element | Config. | Covalent Radius (pm) | IE-1 (kJ/mol) | $\Delta H_{eg}$ (kJ/mol) | Electronegativity |
|---|---|---|---|---|---|
| O | $[He]\ 2s^2\ 2p^4$ | 66 | 1314 | -141 | 3.5 (2nd highest after F) |
| S | $[Ne]\ 3s^2\ 3p^4$ | 104 | 1000 | **-200 (Most negative!)** | 2.5 |
| Se | $[Ar]\ 3d^{10}\ 4s^2\ 4p^4$ | 117 | 941 | -195 | 2.4 |
| Te | $[Kr]\ 4d^{10}\ 5s^2\ 5p^4$ | 137 | 869 | -190 | 2.1 |
| Po | $[Xe]\ 4f^{14}\ 5d^{10}\ 6s^2\ 6p^4$ | 146 | 813 | -174 | 2.0 |

Table 16.1 — Physical and atomic parameters of Group 16 elements.

KEY RESULT
Electron Gain Enthalpy Anomaly:
**S (-200) > Se (-195) > Te (-190) > Po (-174) > O (-141 kJ/mol).**
Oxygen has the LEAST negative electron gain enthalpy in the entire group because of its very compact 2p subshell, causing severe inter-electronic repulsions for incoming electrons!

## 2. Catenation & Allotropy

Sulphur shows maximum catenation in Group 16 because the S-S single bond (213 kJ/mol) is far stronger than the O-O single bond (142 kJ/mol). Polysulphide ions like $S_n^{2-}$ (e.g. $S_2^{2-}$, $S_5^{2-}$) and polythionates exist stably in aqueous solution.

## 3. Multiple Bonding: pπ-pπ vs pπ-dπ

- **Oxygen:** Forms strong **pπ-pπ double bonds** (O=O, C=O, N=O) due to compact 2p orbitals. Hence, oxygen exists as a diatomic gas $O_2$ with weak van der Waals forces.
- **Sulphur & Heavier Chalcogens:** Cannot form stable pπ-pπ bonds; instead, they form single bonds with extensive catenation, leading to polyatomic crown molecules like **$S_8$** (solid at RTP).
- **d-Orbital Participation:** Sulphur expands its octet using vacant 3d orbitals, forming hexa-coordinate molecules like **SF6** and utilizing **pπ-dπ back-bonding** in SO2 and SO3.

**Dioxygen & Ozone** — Paramagnetism of $O_2$ & Strong Oxidising Action of $O_3$

Oxygen exists as triplet ground-state $O_2$ (paramagnetic) and triatomic allotrope Ozone $O_3$ (diamagnetic).

## 1. Dioxygen (O2) — Molecular Orbital Configuration

Valence electron count = 12 electrons. MO configuration:
$\sigma(2s)^2\ \sigma^*(2s)^2\ \sigma(2p_z)^2\ \pi(2p_x)^2\ \pi(2p_y)^2\ \pi^*(2p_x)^1\ \pi^*(2p_y)^1$.
Bond order = (8 - 4)/2 = **2**. It possesses **two unpaired electrons** in degenerate π antibonding orbitals, explaining its experimental **paramagnetism**.

## 2. Ozone (O3) — Synthesis & Angular Resonance Structure

- **Synthesis (Silent Electric Discharge):** $3\ O_2 \rightleftharpoons 2\ O_3$ ($\Delta H° = +142$ kJ/mol, endothermic). A silent electric discharge is used to prevent local heating which would decompose $O_3$ back to $O_2$.
- **Structure:** V-shaped angular geometry ($C_{2v}$ symmetry), $sp^2$ hybridized central O atom, bond angle **117°**, O-O bond length **128 pm** (intermediate between single bond 148 pm and double bond 121 pm due to resonance).
- **Thermodynamics of Ozone:** Thermodynamically unstable: $2\ O_3 \rightarrow 3\ O_2$ ($\Delta H < 0$ and $\Delta S > 0$, so **$\Delta G$ is immensely negative**). High concentrations can decompose explosively!

## 3. Powerful Oxidising Action & Qualitative Tests for Ozone

| Substance Oxidised | Reaction Equation | Observation / Diagnostic Test |
|---|---|---|
| Lead Sulphide (black) | $PbS\ (black) + 4\ O_3 \rightarrow$ **$PbSO_4$ (white)** $+ 4\ O_2$ | Decolourisation of black lead sulphide paper |
| Potassium Iodide (KI) | $2\ KI + H_2O + O_3 \rightarrow$ **$2\ KOH + I_2 + O_2$** | Turns starch paper blue (Quantitative iodometric estimation: titrate $I_2$ with $Na_2S_2O_3$) |
| Mercury Metal (Hg) | **$2\ Hg + O_3 \rightarrow Hg_2O + O_2$** | **Tailing of Mercury:** Hg loses its meniscus and sticks to glass surface due to $Hg_2O$ formation. Restored by shaking with water! |
| Potassium Manganate | $2\ K_2MnO_4$ (green) $+ H_2O + O_3 \rightarrow$ **$2\ KMnO_4$ (purple)** $+ 2\ KOH + O_2$ | Oxidation of green manganate to purple permanganate |

Table 16.2 — Diagnostic oxidising reactions of Ozone.

KEY RESULT
Tailing of Mercury Test: $2\ Hg + O_3 \rightarrow Hg_2O + O_2$. The loss of meniscus is a unique and specific test for Ozone in JEE.

**Allotropes of Sulphur** — Rhombic (α), Monoclinic (β), Chair $S_6$ & Paramagnetic $S_2$

Sulphur exhibits more allotropes than almost any other element. The most important crystalline forms are Rhombic (α) and Monoclinic (β).

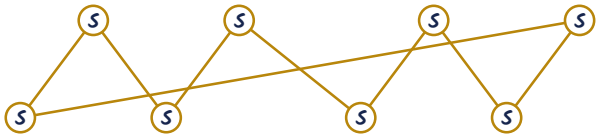
## 1. Rhombic (α-Sulphur) vs Monoclinic (β-Sulphur)

| Property | Rhombic Sulphur (α) | Monoclinic Sulphur (β) |
|---|---|---|
| Crystal System | Orthorhombic (octahedral crystals) | Monoclinic (needle-shaped crystals) |
| Colour & Density | Bright yellow, density 2.06 g/cm³ | Amber yellow, density 1.96 g/cm³ |
| Melting Point | 385.8 K (112.8°C) | 393 K (119.2°C) |
| Stability Range | **Stable below 369 K (96°C)** | **Stable above 369 K (96°C)** |
| Solubility | Insoluble in $H_2O$, soluble in CS2 (≈30 g/100g) | Insoluble in $H_2O$, soluble in CS2 |

Table 16.3 — Comparison of Rhombic and Monoclinic Sulphur.

KEY RESULT
**Transition Temperature (369 K / 95.6°C):** Below 369 K, α-sulphur is stable. Above 369 K, β-sulphur is stable. At exactly 369 K, both allotropes coexist in dynamic equilibrium (Enantiotropic allotropy).

## 2. Molecular Geometry: Crown S8 & Chair S6




**S8 Crown Ring (D4d symmetry)**
**S–S bond length = 204 pm, S–S–S bond angle = 107 deg**

Fig. 16.1 — Puckered $S_8$ crown ring ($D_{4d}$) with bond parameters.

- **Puckered $S_8$ Crown:** Both α and β sulphur consist of puckered $S_8$ crown rings ($D_{4d}$ symmetry). S–S bond distance = 204 pm, S–S–S angle = 107°.
- **Engels Sulphur ($S_6$):** Prepared by reacting $Na_2S_2O_3$ with conc. HCl. Contains a puckered **chair conformation** (like cyclohexane) with S–S–S angle = 102.2° and S–S distance = 205.7 pm.
- **High Temperature $S_2$ Molecule (Paramagnetism):** At elevated temperatures (> 1000 K), sulphur vapor exists predominantly as diatomic **$S_2$** molecules. Just like $O_2$, $S_2$ has **two unpaired electrons** in its π antibonding molecular orbitals and is **paramagnetic**!

## 3. Plastic (Amorphous) Sulphur

Formed by pouring boiling liquid sulphur (at ≈440 K, dark brown viscous liquid with broken $S_8$ chains) in a thin stream into cold water. Yellow-brown rubbery elastic mass containing tangled helical polymer chains ($S_\mu$). It is insoluble in CS2; slowly reverts to rhombic α-sulphur on standing.

**Group 16 Hydrides ($H_2E$)** — Volatility, Acidity, Bond Angles & Drago's Rule

All chalcogens form volatile hydrides $H_2E$ (E = O, S, Se, Te, Po). Properties show sharp trends.

## 1. Periodic Trends in Hydrides

| Property | $H_2O$ | $H_2S$ | $H_2Se$ | $H_2Te$ | Reason / Principle |
|---|---|---|---|---|---|
| Boiling Point (K) | **373** | 213 | 232 | 269 | $H_2O$ is anomalously high due to extensive 3D intermolecular H-bonding; then increases with molar mass (dispersion forces): **$H_2S < H_2Se < H_2Te < H_2O$** |
| Bond Angle (°) | **104.5°** | 92.1° | 91.0° | 90.0° | **Drago's Rule:** In $H_2S$, $H_2Se$, $H_2Te$, no $sp^3$ hybridization occurs; bonds formed by almost pure p orbitals (approx 90°)! |
| Thermal Stability | **High** | Moderate | Low | Very Poor | E-H bond dissociation energy decreases down the group as orbital overlap becomes diffuse: **$H_2O > H_2S > H_2Se > H_2Te$** |
| Reducing Character | Nil | Mild | Strong | **Strongest** | Follows inverse of thermal stability: **$H_2O < H_2S < H_2Se < H_2Te$** |
| Acidic Strength ($K_a$) | $10^{-14}$^ | $10^{-7}$^ | $10^{-4}$^ | **$10^{-3}$^** | E-H bond length increases and bond energy decreases: **$H_2O < H_2S < H_2Se < H_2Te$** |

Table 16.4 — Systematic trends in Group 16 hydrides.

KEY RESULT

Critical Order Sequences:

- **Boiling Point:** $H_2S < H_2Se < H_2Te < \mathbf{H_2O}$
- **Acidic Strength:** $H_2O < H_2S < H_2Se < \mathbf{H_2Te}$
- **Thermal Stability:** $\mathbf{H_2O} > H_2S > H_2Se > H_2Te$
- **Reducing Power:** $H_2O < H_2S < H_2Se < \mathbf{H_2Te}$

## 2. Hydrogen Sulphide (H2S) in Qualitative Analysis

Rotten egg smell, poisonous gas. Precipitates metal cations as insoluble sulphides based on $K_{sp}$:

- **Group II Cations (Acidic medium, dil. HCl):** Low $[S^{2-}]$ due to common ion effect of $H^+$; precipitates low $K_{sp}$ sulphides (CuS, PbS, $Bi_2S_3$, CdS, HgS).
- **Group IV Cations (Basic medium, NH4OH + NH4Cl):** High $[S^{2-}]$ due to removal of $H^+$ by $OH^-$; precipitates higher $K_{sp}$ sulphides (ZnS, NiS, CoS, MnS).

**Oxides of Sulphur** — SO2, SO3 Polymerism & Temporary Bleaching Action

Sulphur forms two commercially crucial oxides: Sulphur dioxide SO2 and Sulphur trioxide SO3.

## 1. Sulphur Dioxide (SO2) — Structure & Bleaching

- **Synthesis:** $S + O_2 \rightarrow SO2$ (industrial) or $Cu + 2\ H_2SO_4$ (conc.) $\xrightarrow{(\Delta)} CuSO4 + SO2\uparrow + 2\ H_2O$ (lab).
- **Structure:** V-shaped angular ($sp^2$ S, one lone pair, O-S-O angle 119.5°, bond length 143 pm equalized by resonance involving pπ-pπ and pπ-dπ bonding).
- **Bleaching Action (Temporary by Reduction):** In the presence of moisture, SO2 acts as a reducing bleaching agent:
  $SO2 + 2\ H_2O \rightarrow H_2SO_4 +$ **2 [H] (Nascent Hydrogen)**
  Coloured organic matter + [H] → **Colourless product**.
  (On exposure to atmospheric oxygen, the bleached matter is re-oxidised back to coloured state. Contrast with $Cl_2$ bleaching, which is permanent by oxidation!).

| Reagent | Reaction with SO2 | Diagnostic Observation |
|---|---|---|
| Acidified $K_2Cr_2O_7$ (orange) | $K_2Cr_2O_7 + H_2SO_4 + 3\ SO2 \rightarrow$ **$Cr_2(SO_4)_3$ (green)** $+ K_2SO_4 + H_2O$ | Orange solution turns emerald green (Test for SO2!) |
| Acidified KMnO4 (purple) | $2\ KMnO4 + 2\ H_2O + 5\ SO2 \rightarrow K_2SO_4 +$ **2 MnSO4 (colorless)** $+ 2\ H_2SO_4$ | Pink/purple permanganate decolourises completely |
| Ferric Chloride (FeCl3) | $2\ FeCl3 + SO2 + 2\ H_2O \rightarrow$ **2 FeCl2 (pale green)** $+ H_2SO_4 + 2\ HCl$ | Yellow ferric solution reduced to green ferrous |
| Iodine solution ($I_2$) | $I_2 + SO2 + 2\ H_2O \rightarrow$ **2 HI (colorless)** $+ H_2SO_4$ | Brown iodine solution decolourises |

Table 16.5 — Characteristic redox reactions of Sulphur Dioxide.

## 2. Sulphur Trioxide (SO3) & Solid Allotropes

In the gas phase, SO3 is planar trigonal ($sp^2$, $D_{3h}$ symmetry, three equivalent S=O bonds). In the solid state, it undergoes remarkable polymerisation:
· **γ-SO3 (Trimer):** Cyclic trimer **$(SO3)3$** or $S_3O_9$ with a puckered 6-membered ring of alternating S and O atoms.
· **β-SO3 & α-SO3 (Linear Polymers):** Infinite helical chains of tetrahedral SO4 units sharing corners, resembling asbestos fibers (mp 32°C and 62°C).

**Oxoacids of Sulphur (Part I)** — Contact Process & $H_2SO_4$ Dehydrating Action

Sulphuric acid $H_2SO_4$ ('King of Chemicals') is manufactured worldwide via the Contact Process.

## 1. Contact Process for H2SO4 Manufacture

- **Step 1:** Burning sulphur or iron pyrites in air: $S + O_2 \rightarrow SO2$ (or $4\ FeS2 + 11\ O_2 \rightarrow 2\ Fe_2O_3 + 8\ SO2$).
- **Step 2 (The Catalytic Key Step):** Catalytic oxidation of SO2 to SO3:
  **$2\ SO2(g) + O_2(g) \leftarrow(V_2O_5,\ 720\ K,\ 2\ bar)\rightarrow 2\ SO3(g)$** $\Delta H° = -196.6$ kJ/mol (Exothermic).
  (Optimal conditions: excess $O_2$, 2 bar pressure, 720 K with Vanadium pentoxide $V_2O_5$ catalyst).
- **Step 3:** Absorption of SO3 in 98% $H_2SO_4$ to form **Oleum (Pyrosulphuric acid $H_2S_2O_7$)**:
  $SO3 + H_2SO_4 \rightarrow$ **$H_2S_2O_7$**.
  (Direct absorption in water is avoided because it forms a dense, unmanageable acid mist!).
- **Step 4:** Dilution of oleum with calculated quantity of water: $H_2S_2O_7 + H_2O \rightarrow$ **$2\ H_2SO_4$**.

## 2. Chemical Properties & Dehydrating Action of H2SO4

| Reaction Category | Substrate | Reaction Equation & Observation |
|---|---|---|
| Dehydrating Action | Cane Sugar (Sucrose $C_{12}H_{22}O_{11}$) | $C_{12}H_{22}O_{11}$ —(conc. $H_2SO_4$)→ **12 C (black spongy char) + 11 $H_2O$** (Charring of sugar) |
| Dehydrating Action | Ethanol ($C_2H_5OH$) | $C_2H_5OH$ —(conc. $H_2SO_4$, 443 K)→ **$C_2H_4$ (Ethene)** + $H_2O$ |
| Dehydrating Action | Formic Acid (HCOOH) | $HCOOH$ —(conc. $H_2SO_4$)→ **CO**↑ + $H_2O$ |
| Dehydrating Action | Oxalic Acid ((COOH)2) | $(COOH)2$ —(conc. $H_2SO_4$)→ **CO**↑ **+ CO2**↑ + $H_2O$ |
| Oxidising Action | Copper (Cu) | $Cu + 2\ H_2SO_4$ (conc.) —(Δ)→ $CuSO4$ + **SO2**↑ + $2\ H_2O$ |
| Oxidising Action | Carbon (C) & Sulphur (S) | $C + 2\ H_2SO_4 \rightarrow$ **CO2 + 2 SO2** + $2\ H_2O$; $S + 2\ H_2SO_4 \rightarrow$ **3 SO2** + $2\ H_2O$ |

Table 16.6 — Diagnostic dehydrating and oxidising reactions of Sulphuric Acid.

KEY RESULT
Dilution Rule: Always add **acid slowly to water with constant stirring**, NEVER water to concentrated acid (huge exothermic hydration causes dangerous boiling and acid splattering!).

**Oxoacids of Sulphur (Part II)** — Peroxoacids, Thionic Series & Thiosulphate

Peroxoacids contain the -O-O- peroxo linkage (formal oxidation state of peroxo oxygens = -1).

## 1. Peroxomonosulphuric Acid (Caro's Acid, H2SO5)

Prepared by reacting chlorosulphonic acid with hydrogen peroxide: ClSO3H + $H_2O_2$ → **$H_2SO_5$** + HCl.
Structural Formula: **HO-SO2-O-O-H**.
Formal Oxidation State Calculation: 2(+1) + S + 3(-2) + 2(-1) = 0 → **S = +6** (NOT +8!).
Contains one peroxo (-O-O-) bond, two S=O bonds, and one normal S-OH bond.

## 2. Peroxodisulphuric Acid (Marshall's Acid, H2S2O8)

Prepared by anodic oxidation of cold 50% $H_2SO_4$ at high current density: 2 $HSO_4^-$ → **$H_2S_2O_8$** + $2e^-$.
Structural Formula: **HO-SO2-O-O-SO2-OH**.
Oxidation State of both Sulphur atoms = **+6** (contains one bridging -O-O- peroxide bond).
Hydrolysis: $H_2S_2O_8 + H_2O \rightarrow H_2SO_5 + H_2SO_4$; $H_2SO_5 + H_2O \rightarrow H_2SO_4 + \mathbf{H_2O_2}$ (industrial route to $H_2O_2$).

| Oxoacid | Formula | S Oxidation State | Key Structural Linkage | Basicity |
|---|---|---|---|---|
| Sulphurous acid | $H_2SO_3$ | +4 | One lone pair on S, one S=O, two S-OH | 2 (Dibasic) |
| Dithionous acid | $H_2S_2O_4$ | +3 | Direct **S-S bond**, two S=O, two S-OH | 2 (Dibasic) |
| Dithionic acid | $H_2S_2O_6$ | +5 | Direct **S-S bond**, four S=O, two S-OH | 2 (Dibasic) |
| Polythionic acid | $H_2S_nO_6$ | Average | Polysulphide chain **$-S_{n-2}-$** flanked by -SO3H | 2 (Dibasic) |
| Pyrosulphuric (Oleum) | $H_2S_2O_7$ | +6 | **S-O-S bridge**, four S=O, two S-OH | 2 (Dibasic) |
| Caro's acid | $H_2SO_5$ | +6 | **One -O-O- bridge**, two S=O, one S-OH | 1 (Monobasic!) |
| Marshall's acid | $H_2S_2O_8$ | +6 | **One -O-O- bridge**, four S=O, two S-OH | 2 (Dibasic) |

Table 16.7 — Master classification of Sulphur oxoacids.

## 3. Sodium Thiosulphate (Na2S2O3·5H2O, 'Hypo')

- **Synthesis:** $Na_2SO_3 + S \xrightarrow{\text{(boil with water)}} Na_2S_2O_3$.
  Structure: Tetrahedral with one O replaced by S: central S is +5, terminal S is -1 (Average = +2).
- **Photography (Fixer):** AgBr + 2 $Na_2S_2O_3$ → **$Na_3[Ag(S_2O_3)_2]$ (soluble complex)** + NaBr.
- **Iodometry:** 2 $Na_2S_2O_3 + I_2$ → **$Na_2S_4O_6$ (Sodium tetrathionate)** + 2 NaI.
- **With Strong Oxidant ($Cl_2$):** $Na_2S_2O_3 + 4\ Cl_2 + 5\ H_2O \rightarrow$ **2 NaHSO4 + 8 HCl + $H_2SO_4$** (acts as 'Antichlor').

**Sulphur Halides & Oxohalides** — SF4 (See-Saw), SF6 (Inert Octahedron) & Thionyl Chloride

Sulphur forms hexafluoride SF6, tetrahalides SF4, SCl4, and dihalides SCl2, $S_2Cl_2$.

## 1. Sulphur Hexafluoride (SF6) — Kinetic Inertness

- **Structure:** Regular Octahedral ($sp^3d^2$, $O_h$ symmetry, all S-F bonds = 156 pm).
- **Extraordinary Chemical Inertness:** SF6 is non-toxic, odorless, completely inert to water, boiling molten KOH, and acids. Does not hydrolyse even at 500°C!
- **Reason for Inertness:** S atom is coordinatively saturated and sterically shielded by six compact Fluorine atoms. Water molecules cannot approach the central Sulphur atom, and Sulphur cannot expand its coordination sphere further.
- **Applications:** High dielectric strength gas used as an electrical insulator in high-voltage circuit breakers and transformers.

| Compound | Hybridization | Geometry (VSEPR) | Hydrolysis & Reactivity |
|---|---|---|---|
| SF4 | $sp^3d$ (1 lone pair eq) | **See-Saw (Distorted tetrahedron)** | Hydrolyses violently: SF4 + 2 $H_2O$ → **SO2 + 4 HF** (fluorinating agent) |
| SF6 | $sp^3d^2$ (0 lone pair) | **Regular Octahedral** | Completely inert to water and hot caustic alkali |
| SCl4 | $sp^3d$ (1 lone pair) | See-Saw | Thermally unstable; decomposes above -30°C: SCl4 → SCl2 + $Cl_2$ |
| $S_2Cl_2$ | $sp^3$ (bent per S) | Open-book / $H_2O_2$-like | Yellow liquid; reacts with water: 2 $S_2Cl_2$ + 2 $H_2O$ → SO2 + 3 S + 4 HCl |

Table 16.8 — Comparative structures and reactions of Sulphur halides.

## 2. Thionyl Chloride (SOCl2) & Sulphuryl Chloride (SO2Cl2)

- **Thionyl Chloride (SOCl2):** Pyramidal ($sp^3$, 1 lone pair, one S=O, two S-Cl).
  Darzens reaction (organic synthesis of alkyl chlorides): R-OH + SOCl2 —(pyridine)→ **R-Cl + SO2↑ + HCl↑**.
  (Both by-products SO2 and HCl are gases, escaping spontaneously to leave pure alkyl chloride!).
- **Sulphuryl Chloride (SO2Cl2):** Tetrahedral ($sp^3$, two S=O, two S-Cl).
  Hydrolysis: SO2Cl2 + 2 $H_2O$ → **$H_2SO_4$ + 2 HCl**.

**Group 17: Halogens ($ns^2\ np^5$)** — Salt Formers & Periodic Anomalies

Contains F, Cl (gases), Br (volatile red liquid), I (lustrous solid), At, Ts (radioactive). Highest effective nuclear charge in their respective periods, having the most negative electron gain enthalpies.

## 1. Periodic Trends of Halogens

| Element | Config. | Covalent Radius (pm) | IE-1 (kJ/mol) | $\Delta H_{eg}$ (kJ/mol) | Bond Enthalpy $X_2$ (kJ/mol) |
|---|---|---|---|---|---|
| F | [He] $2s^2\ 2p^5$ | 64 | 1680 | -328 | **158.8 (Low!)** |
| Cl | [Ne] $3s^2\ 3p^5$ | 99 | 1256 | **-349 (Most negative!)** | **242.6 (Highest!)** |
| Br | [Ar] $3d^{10}\ 4s^2\ 4p^5$ | 114 | 1142 | -325 | 192.8 |
| I | [Kr] $4d^{10}\ 5s^2\ 5p^5$ | 133 | 1008 | -295 | 151.1 |

Table 17.1 — Physical and atomic properties of Group 17 elements.

KEY RESULT

Bond Dissociation Enthalpy Anomaly (Classic JEE Question):
**$Cl_2$ (242.6) > $Br_2$ (192.8) > $F_2$ (158.8) > $I_2$ (151.1 kJ/mol).**
$F_2$ has a lower bond energy than both $Cl_2$ and $Br_2$ because of severe inter-electronic repulsions between the non-bonding electron pairs in the extremely small $F_2$ molecule (F-F distance = 143 pm)!

## 2. Oxidising Power & Electrode Potentials

Standard reduction potential: $F_2$ (+2.87 V) > $Cl_2$ (+1.36 V) > $Br_2$ (+1.09 V) > $I_2$ (+0.54 V).
Fluorine is the most powerful oxidising agent in chemistry. Why $F_2$ > $Cl_2$ despite lower electron gain enthalpy? Standard reduction potential depends on the Born-Haber cycle parameters:

- 1. **Low Bond Dissociation Enthalpy:** $\Delta H_{diss}(F_2)$ is very low (+158.8 kJ/mol) → easy atomisation.
- 2. **Very High Hydration Enthalpy:** $F^-$ is small with immense charge density, releasing huge hydration enthalpy: $\Delta H_{hyd}(F^-)$ = **-515 kJ/mol** (vs $Cl^-$ = -381 kJ/mol).
- 3. **Conclusion:** High hydration energy and low bond dissociation energy more than compensate for the slightly less negative electron gain enthalpy of $F_2$!

## 3. Anomalous Behaviour of Fluorine

Only oxidation state is **-1** (no d-orbitals, most electronegative element). Maximum covalency is strictly 1 (or 2 in rare bridging/coordination). HF is a weak acid with high boiling point (293 K) due to zig-zag hydrogen-bonded chains (H-F...H-F...).

**Extraction of Halogens** — Moissan Electrolysis, Deacon Process & $Br_2/I_2$ Mining

Because $F_2$ is the strongest oxidant, no chemical reagent can oxidise $F^-$ to $F_2$; it is produced exclusively by electrolysis.

## 1. Extraction of Fluorine (Dennis-Moissan Process)

- **Electrolyte:** Anhydrous molten mixture of **KF + 2 HF (KF·2HF)** at ≈100°C in a Monel metal or steel vessel.
- **Why Anhydrous HF alone cannot be used:** Pure anhydrous liquid HF is a non-electrolyte (does not conduct electricity). Adding KF supplies $K^+$ and $[HF2]^-$ ions, making it conductive.
- **Electrodes:** Carbon (graphite) anode, steel vessel as cathode. A copper diaphragm separates anode and cathode to prevent explosive recombination of $H_2$ and $F_2$:
  - At Cathode: $2\ H^+ + 2e^- \rightarrow \mathbf{H_2}\uparrow$
  - At Anode: $2\ F^- \rightarrow \mathbf{F_2}\uparrow + 2e^-$

## 2. Manufacture of Chlorine (Cl2)

| Process | Reagents / Mechanism | Key Reaction Equation |
|---|---|---|
| Deacon's Process | Oxidation of HCl gas by atmospheric $O_2$ with CuCl2 catalyst (723 K) | $4\ HCl + O_2 \xrightarrow{CuCl2,\ 723\ K} \mathbf{2\ Cl_2 + 2\ H_2O}$ |
| Chlor-Alkali (Nelson / Castner-Kellner) | Electrolysis of aqueous brine (NaCl) | $2\ NaCl + 2\ H_2O \xrightarrow{electrolysis} \mathbf{Cl_2\ (anode) + H_2\ (cathode) + 2\ NaOH}$ |
| Laboratory Prep | Oxidation of conc. HCl with MnO2 or KMnO4 | $MnO2 + 4\ HCl \rightarrow \mathbf{MnCl2 + Cl_2}\uparrow \mathbf{+ 2\ H_2O}$<br>$2\ KMnO4 + 16\ HCl \rightarrow \mathbf{2\ KCl + 2\ MnCl2 + 5\ Cl_2}\uparrow \mathbf{+ 8\ H_2O}$ |

Table 17.2 — Synthetic and industrial processes for Chlorine.

## 3. Recovery of Bromine and Iodine

- **Bromine from Sea Brine:** Sea water contains $Br^-$ (≈65 ppm). Acidified with $H_2SO_4$, treated with $Cl_2$ gas: $\mathbf{2\ Br^- + Cl_2 \rightarrow 2\ Cl^- + Br_2}\uparrow$. Blown out with air, absorbed in $Na_2CO_3$ solution: $3\ Na_2CO_3 + 3\ Br_2 \rightarrow 5\ NaBr + NaBrO3 + 3\ CO2$.
- **Iodine from Caliche (Chile Saltpetre):** Contains sodium iodate NaIO3. Reduced with sodium bisulphite: $\mathbf{2\ NaIO3 + 5\ NaHSO3 \rightarrow 2\ Na_2SO_4 + 3\ NaHSO4 + I_2}\downarrow \mathbf{+ H_2O}$.
- **Iodine from Seaweeds (Laminaria):** Ash (Kelp) contains iodides; boiled with conc. $H_2SO_4$ and MnO2: $2\ NaI + MnO2 + 3\ H_2SO_4 \rightarrow 2\ NaHSO4 + MnSO4 + \mathbf{I_2}\uparrow + 2\ H_2O$.

**Hydrogen Halides (HX)** — Boiling Points, Bond Energies, Acidity & Reducing Trends

Prepared by direct combination or by salt + acid reactions: $NaX + H_2SO_4 \rightarrow NaHSO4 + HX$.

## 1. Periodic Trends in Hydrogen Halides

| Property | HF | HCl | HBr | HI | Trend / Underlying Reason |
|---|---|---|---|---|---|
| Boiling Point (K) | **293 (Highest!)** | 189 | 206 | 238 | HF is anomalously high due to strong intermolecular H-bonding; then increases with van der Waals dispersion forces: **HCl < HBr < HI < HF** |
| Bond Length (pm) | 91.7 | 127.4 | 141.4 | **160.9** | Increases down group with increasing halogen size |
| Bond Enthalpy (kJ/mol) | **574** | 432 | 363 | 295 | Decreases sharply as overlap efficiency declines |
| Thermal Stability | **Most Stable** | Stable | Decomposes >800 K | **Decomposes readily** | Parallels bond dissociation energy: **HF > HCl > HBr > HI** |
| Acidic Strength ($K_a$) | Weak ($3.5\times10^{-4}$) | Strong ($\approx10^{7}$) | Very Strong ($\approx10^{9}$) | **Strongest ($\approx10^{10}$)** | Follows inverse of H-X bond strength: **HF < HCl < HBr < HI** |
| Reducing Character | Nil | Very Weak | Strong | **Strongest** | HI easily oxidises to $I_2$: **HF < HCl < HBr < HI** |

Table 17.3 — Master summary of Hydrogen Halide properties.

KEY RESULT

Critical Sequences for JEE:

- **Boiling Point:** HCl < HBr < HI < **HF**
- **Acidic Strength:** HF < HCl < HBr < **HI**
- **Reducing Power:** HF < HCl < HBr < **HI**
- **Thermal Stability:** **HF** > HCl > HBr > HI

## 2. Preparation Traps with H2SO4

TRAP

TRAP: HBr and HI CANNOT be prepared by heating NaBr/NaI with conc. $H_2SO_4$!

Conc. $H_2SO_4$ is a strong oxidising agent that oxidises HBr to $Br_2$ and HI to $I_2$:

$2\ NaBr + 2\ H_2SO_4 \rightarrow Na_2SO_4 + \mathbf{Br_2}$ **(red vapor)** $+ SO2 + 2\ H_2O$

$2\ NaI + 2\ H_2SO_4 \rightarrow Na_2SO_4 + \mathbf{I_2}$ **(violet vapor)** $+ SO2 + 2\ H_2O$.

**Correct Method:** Use non-oxidising **conc. $H_3PO_4$**: $3\ NaBr + H_3PO_4 \xrightarrow{(\Delta)} Na_3PO_4 + 3\ HBr\uparrow$.

**Halogen Oxides** — OF2, Paramagnetic ClO2, Dimeric $Cl_2O_6$ & $I_2O_5$ in CO Estimation

Fluorine forms oxygen fluorides (F is more electronegative than O). Other halogens form oxides.

## 1. Oxygen Fluorides (OF2 and O2F2)

- **OF2 (Oxygen Difluoride):** Prepared: 2 NaOH (cold, 2% dil.) + 2 $F_2$ → **OF2**↑ + 2 NaF + $H_2O$. V-shaped (angle 103°, F-O-F repulsions compress angle below water's 104.5°). Oxidation state of O is **+2**.
- **$O_2F_2$ (Dioxygen Difluoride):** Prepared: $F_2$ + $O_2$ in electric discharge at 90 K. Open-book structure like $H_2O_2$. Oxidation state of O is **+1**. Extremely powerful fluorinating agent; reacts with plutonium: **Pu + 3 $O_2$ $F_2$ → PuF6 + 3 $O_2$** (used to recover Pu from spent nuclear fuel!).

## 2. Oxides of Chlorine

| Oxide | Oxid. State | Structure & Geometry | Key Reaction / Diagnostic Behavior |
|---|---|---|---|
| $Cl_2O$ | +1 | Bent V-shaped (angle 111°) | 2 HgO + 2 $Cl_2$ → HgO·HgCl2 + **$Cl_2O$**; Anhydride of HOCl |
| ClO2 | +4 | Bent (odd-electron, 19 $e^-$) | **Paramagnetic gas**; does not dimerise readily; bleaching agent for wood pulp; 2 ClO2 + 2 NaOH → **NaClO2 + NaClO3** + $H_2O$ (mixed anhydride!) |
| $Cl_2O_6$ | +6 | **Liquid / Solid: $[ClO_2]^+$ $[ClO_4]^-$** | Mixed anhydride of HClO3 and HClO4: $Cl_2O_6$ + $H_2O$ → **HClO3 + HClO4** |
| $Cl_2O_7$ | +7 | Two ClO3 tetrahedra sharing O bridge | Anhydride of perchloric acid: 2 HClO4 + $P_4O_{10}$ → **$Cl_2O_7$** + 4 HPO3. Colorless oily liquid, explodes violently on percussion! |

Table 17.4 — Comprehensive spectrum of Chlorine Oxides.

## 3. Iodine Pentoxide (I2O5) & Quantitative Estimation of CO

KEY RESULT

Prepared by dehydrating iodic acid: 2 HIO3 —(440 K)→ **$I_2O_5$** + $H_2O$.
White crystalline solid, highly stable. Reacts quantitatively with Carbon Monoxide at 170°C:
**$I_2O_5$ + 5 CO → $I_2$ + 5 CO2**.
The liberated $I_2$ is titrated against standard sodium thiosulphate ($Na_2S_2O_3$) with starch indicator. This is the standard analytical method for estimating toxic CO in air and flue gases!

**Oxoacids of Halogens** — Acid Strengths, Anion Resonance & Thermal Stability

Halogens form four series of oxoacids: Hypohalous (HOX, +1), Halous (HXO2, +3), Halic (HXO3, +5), Perhalic (HXO4, +7).

## 1. Complete Classification of Halogen Oxoacids

| Oxidation State | Acid Family | Fluorine | Chlorine | Bromine | Iodine |
|---|---|---|---|---|---|
| +1 | Hypohalous (HOX) | HOF (only one!) | HOCl | HOBr | HOI |
| +3 | Halous (HXO2) | — | HClO2 | — | — |
| +5 | Halic (HXO3) | — | HClO3 | HBrO3 | HIO3 |
| +7 | Perhalic (HXO4) | — | HClO4 | HBrO4 | HIO4 / $H_5IO_6$ (Periodic) |

Table 17.5 — Systematic matrix of halogen oxoacids.

## 2. Acid Strength Trends

- **Trend 1 (Same Halogen, Increasing Oxidation State):**
  **HOCl (+1) < HClO2 (+3) < HClO3 (+5) < HClO4 (+7).**
  Reason: Conjugate base stability increases as the negative charge is delocalised over more electronegative oxygen atoms via resonance: $ClO^-$ (1 O) < $ClO_2^-$ (2 O) < $ClO_3^-$ (3 O) < **$ClO_4^-$ (4 O, charge density −0.25/O)**. HClO4 is one of the strongest known Bronsted acids!
- **Trend 2 (Different Halogens, Same Oxidation State):**
  **HOCl > HOBr > HOI.**
  Reason: As electronegativity of halogen decreases (Cl > Br > I), polarity of the O-H bond decreases, making release of $H^+$ less favorable.

## 3. Oxidising Power vs Thermal Stability

| Family Sequence | Acid Strength Order | Thermal Stability Order | Oxidising Power Order |
|---|---|---|---|
| Chlorine Series | HOCl < HClO2 < HClO3 < **HClO4** | HOCl < HClO2 < HClO3 < **HClO4** | **HOCl** > HClO2 > HClO3 > HClO4 |

Table 17.6 — Inverse relationship between stability and oxidising power.

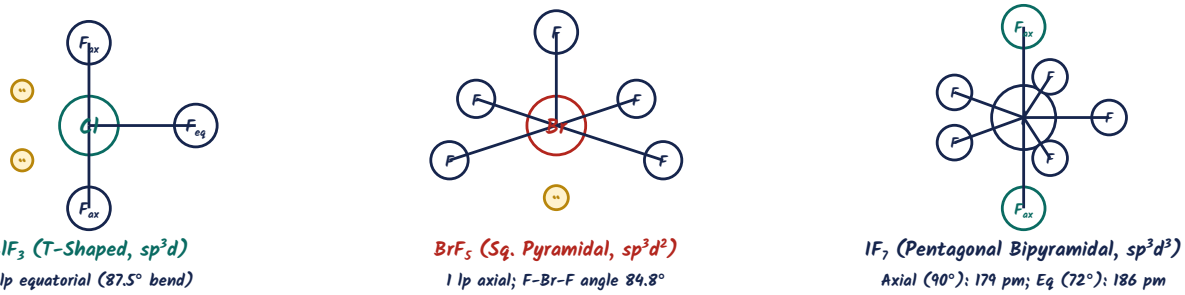
KEY RESULT

**Oxidising Power is INVERSE to acid stability:** Hypochlorous acid **HOCl is the STRONGEST oxidant**, whereas perchloric acid **HClO4 is the weakest oxidant** in dilute aqueous solution!

**Interhalogen Compounds ($XX'_n$)** — Synthesis, VSEPR Geometries & Reactivity

Formed when two different halogens react. General formula $XX'_n$ where X is the larger halogen and X' is the smaller, more electronegative halogen; n = 1, 3, 5, 7.

## 1. Comprehensive VSEPR Geometries of Interhalogens




$ClF_3$ (T-Shaped, $sp^3d$)
2 lp equatorial (87.5° bend)

$BrF_5$ (Sq. Pyramidal, $sp^3d^2$)
1 lp axial; F-Br-F angle 84.8°

$IF_7$ (Pentagonal Bipyramidal, $sp^3d^3$)
Axial (90°): 179 pm; Eq (72°): 186 pm

Fig. 17.1 — Interhalogen geometries: ClF3 (T-Shaped), BrF5 (Square Pyramidal), and IF7 (Pentagonal Bipyramidal).

| Type | Example | Hybridization | Lone Pairs on X | VSEPR Geometry | Physical State |
|---|---|---|---|---|---|
| XX' | ClF, BrF, ICl | $sp^3$ | 3 lone pairs | **Linear** | ClF (gas), ICl (ruby red solid) |
| $XX'_3$ | ClF3, BrF3, ICl3 | $sp^3d$ | 2 equatorial lp | **T-Shaped** (bent T) | ClF3 (gas), BrF3 (liquid), ICl3 (dimer solid) |
| $XX'_5$ | ClF5, BrF5, IF5 | $sp^3d2$ | 1 axial lp | **Square Pyramidal** | All colorless liquids/gases |
| $XX'_7$ | **IF7** (only one!) | $sp^3d3$ | 0 lone pairs | **Pentagonal Bipyramidal** | Colorless gas ($D_{5h}$ symmetry) |

Table 17.7 — Complete structural geometries of interhalogens.

KEY RESULT
Why is IF7 the ONLY $XX'_7$ compound formed?
Because Iodine has the largest atomic radius in Group 17, and Fluorine has the smallest atomic radius. The radius ratio r(I)/r(F) is maximum, allowing sterically uncrowded coordination of seven F atoms!

## 2. Chemical Reactivity: Interhalogens vs Halogens

In general, **interhalogen compounds are MORE reactive than pure halogens (except $F_2$)**.
Reason: The X-X' bond in an interhalogen is polar and weaker than the non-polar X-X bond in a pure halogen (except F-F where lone pair repulsions weaken the bond).

## 3. Hydrolysis of Interhalogens

Hydrolysis always yields a **halide ion from the smaller halogen X'** and an **oxohalate ion from the larger central halogen X**:

- $XX' + H_2O \rightarrow HX' + HOX$ (e.g., $ICl + H_2O \rightarrow HCl + HOI$)
- $XX'_3 + 2\ H_2O \rightarrow 3\ HX' + HXO2$ (e.g., $BrF3 + 2\ H_2O \rightarrow 3\ HF + HBrO2$)
- $XX'_5 + 3\ H_2O \rightarrow 5\ HX' + HXO3$ (e.g., $IF5 + 3\ H_2O \rightarrow 5\ HF + HIO3$)
- $XX'_7 + 4\ H_2O \rightarrow 7\ HX' + HXO4$ (e.g., $IF7 + 4\ H_2O \rightarrow 7\ HF + H_5IO_6$)

TRAP
TRAP: ICl3 in the solid state exists as a planar bridged dimer $I_2Cl_6$ with two bridging chlorine atoms (chlorine-bridged like $Al_2Cl_6$)!

**Polyhalide Ions** — Triiodide $I_3^-$, $ICl_2^-$, $ICl_4^-$ & 3c-4e Hypervalent Bonding

Halogens and interhalogens coordinate with halide ions to form polyhalide anions: $X^- + n\,X_2 \rightarrow X_{2n+1}^-$.

## 1. Triiodide Ion (I3-) — Structure & Bonding

- **Solubility of $I_2$ in KI:** Pure iodine is virtually insoluble in pure water (non-polar). It dissolves readily in aqueous KI due to formation of the soluble **triiodide complex anion**:
  **$KI + I_2 \rightleftharpoons K^+[I_3]^-$**.
- **VSEPR Geometry:** Central I atom has 7 valence $e^-$ + 1 from charge + 2 from bond pairs = 10 electrons = 5 pairs ($sp^3d$ hybridization).
- **Three Equatorial Lone Pairs:** The three lone pairs occupy equatorial positions at 120° to minimise repulsions. The two I-I bonds are axial (180°).
  Geometry is **perfectly LINEAR ($D_{\infty h}$ symmetry)** with I-I bond length 292 pm.

| Polyhalide Ion | Central Atom | Hybridization | Lone Pairs | Geometry (VSEPR) | Bond Angle |
|---|---|---|---|---|---|
| $I_3^-$ | I | $sp^3d$ | 3 equatorial | **Linear** | 180° |
| $ICl_2^-$ | I | $sp^3d$ | 3 equatorial | **Linear** | 180° |
| $ICl_4^-$ | I | $sp^{3d}2$ | 2 axial | **Square Planar** | 90° |
| $I_3^+$ (Cation) | I | $sp^3$ | 2 lone pairs | **Bent / V-shaped** | 104° |
| $I_5^-$ | I | — | — | **V-shaped dimer** ($I_3^-$ coordinated to $I_2$) | ≈95° |

Table 17.8 — Geometries and electron pairs of Polyhalide species.

KEY RESULT
Why does Fluorine NOT form polyhalide anions ($F_3^-$ does not exist)?
Because Fluorine has no vacant d orbitals to expand its octet to accommodate 10 valence electrons ($sp^3d$)!

## 2. Polyhalide Cations

Strong oxidising Lewis acids like AsFS or SbFS can pull $F^-$ from interhalogens to generate polyhalide cations:
- $ClF_3 + SbFS \rightarrow$ **$[ClF_2]^+[SbF_6]^-$** (bent $[ClF_2]^+$, $sp^3$, 2 lone pairs)
- $BrFS + SbFS \rightarrow$ **$[BrF_4]^+[SbF_6]^-$** (see-saw $[BrF_4]^+$, $sp^3d$, 1 lone pair)

**Pseudohalogens & Pseudohalides** — Cyanogen (CN)2, Thiocyanogen (SCN)2 & Chemistry

Pseudohalogens are univalent polyatomic groupings that resemble true halogens ($X_2$) in their free state, forming univalent anions (pseudohalides $X^-$) and volatile hydrogen acids (HX).

## 1. Defining Criteria for Pseudohalogens

- 1. They consist of two or more electronegative atoms, at least one of which is Nitrogen.
- 2. They dimerise in the free state to neutral symmetric volatile dimers: **(CN)2 (Cyanogen)**, **(SCN)2 (Thiocyanogen)**, **(SeCN)2**, **(OCN)2**.
- 3. They react with alkalis to disproportionate into pseudohalide and pseudohalite (like $Cl_2 + 2\ OH^- \rightarrow Cl^- + ClO^- + H_2O$):
  **$(CN)2 + 2\ OH^- \rightarrow CN^-$ (Cyanide) + $OCN^-$ (Cyanate) + $H_2O$.**
- 4. They form insoluble silver, lead, and mercury(I) salts: AgCN (white↓), AgSCN (white↓), resembling AgCl↓.

| Pseudohalide Ion | Formula | Dimeric Pseudohalogen | Insoluble $Ag^+$ Salt & Behaviour in NH3 |
|---|---|---|---|
| Cyanide | $CN^-$ | (CN)2 (Cyanogen, gas) | AgCN↓ (white, dissolves in excess NH3 and in excess $CN^-$) |
| Thiocyanate | $SCN^-$ | (SCN)2 (Thiocyanogen, yellow oil) | AgSCN↓ (white, insoluble in dil. HNO3; reacts with $Fe^{3+}$ → blood-red $[Fe(SCN)]^{2+}$) |
| Cyanate | $OCN^-$ | (OCN)2 | AgOCN↓ (white) |
| Azide | $N_3^-$ | $(N_3)_2$ (Does NOT exist!) | AgN3↓ (white, shock-sensitive explosive detonator) |

Table 17.9 — Classification of Pseudohalogens and Pseudohalides.

KEY RESULT
TRAP Question: Is Azide ($N_3^-$) a true pseudohalide?
$N_3^-$ is an **incomplete pseudohalide** because the free neutral dimer '$(N_3)_2$' (hexanitrogen $N_6$) does NOT exist as a stable isolable molecule!

**Bleaching Powder & Disproportionation** — CaOCl2, Available Chlorine & Alkali Reactions

Bleaching powder is manufactured by action of chlorine on dry slaked lime: $Ca(OH)_2 + Cl_2 \rightarrow CaOCl2{\cdot}H_2O$.

## 1. Composition & Chemical Nature of Bleaching Powder

- **True Composition:** Bleaching powder is NOT a simple single compound CaOCl2. Its actual structural formula is a mixed double salt: $\mathbf{Ca(OCl)_2{\cdot}CaCl_2{\cdot}Ca(OH)_2{\cdot}2H_2O}$.
- **Oxidation State of Chlorine in CaOCl2:**
  CaOCl2 dissociates into $Ca^{2+}$, $Cl^-$, and $OCl^-$ ions.
  · Direct Chloride ion ($Cl^-$): **Oxidation state = −1**
  · Hypochlorite ion ($OCl^-$): **Oxidation state = +1**
  · Average oxidation state = **0**! (Classic JEE question).
- **Auto-Decomposition on Long Standing:** 6 CaOCl2 —(standing in air)→ $\mathbf{Ca(ClO_3)_2}$ **+ 5 CaCl2** (loss of bleaching power).

## 2. Available Chlorine & Iodometric Titration

The percentage of active chlorine liberated by treatment with excess dilute acid is called **Available Chlorine**:
CaOCl2 + 2 CH3COOH (or dil. $H_2SO_4$) → $Ca(CH_3COO)_2 + H_2O$ + $\mathbf{Cl_2}\uparrow$.
The liberated $Cl_2$ is reacted with excess KI: $Cl_2 + 2\ KI \rightarrow 2\ KCl + \mathbf{I_2}$. The liberated $I_2$ is titrated against standard 0.1 N $Na_2S_2O_3$ with starch indicator: $\mathbf{I_2 + 2\ Na_2S_2O_3 \rightarrow 2\ NaI + Na_2S_4O_6}$.

## 3. Master Summary of Halogen Disproportionation with Alkali

| Halogen | With Cold, Dilute NaOH (15°C) | With Hot, Concentrated NaOH (70°C) |
|---|---|---|
| $Cl_2$ | $Cl_2 + 2\ NaOH \rightarrow$ **NaCl (−1) + NaOCl (+1)** + $H_2O$ | $3\ Cl_2 + 6\ NaOH \rightarrow$ **5 NaCl (−1) + NaClO3 (+5)** + $3\ H_2O$ |
| $Br_2$ | $Br_2 + 2\ NaOH \rightarrow$ **NaBr + NaOBr** + $H_2O$ (unstable) | $3\ Br_2 + 6\ NaOH \rightarrow$ **5 NaBr (−1) + NaBrO3 (+5)** + $3\ H_2O$ |
| $I_2$ | Disproportionates instantly to iodate even in cold! | $3\ I_2 + 6\ NaOH \rightarrow$ **5 NaI (−1) + NaIO3 (+5)** + $3\ H_2O$ |
| $F_2$ (Redox!) | $\mathbf{2\ F_2 + 2\ NaOH\ (dil) \rightarrow 2\ NaF + OF2\uparrow + H_2O}$ | $\mathbf{2\ F_2 + 4\ NaOH\ (conc) \rightarrow 4\ NaF + O_2\uparrow + 2\ H_2O}$ |

Table 17.10 — Reaction of Halogens with Cold/Dilute vs Hot/Concentrated Alkali.

KEY RESULT
Fluorine does NOT disproportionate with alkali! Because F cannot have positive oxidation states, it oxidises hydroxide oxygen: **cold dil. NaOH gives OF2**, and **hot conc. NaOH gives $O_2$**!

**Group 18: Noble Gases ($ns^2\ np^6$)** — Periodic Trends, Liquid Helium & Clathrates

Contains He, Ne, Ar, Kr, Xe, Rn (radioactive). Completely filled valence shells confer extraordinary chemical inertness, extremely high ionization energies, and zero electron affinity.

## 1. Periodic Trends in Group 18

| Element | Config. | Atomic Radius (pm) | IE-1 (kJ/mol) | $\Delta H_{eg}$ (kJ/mol) | Boiling Point (K) |
|---|---|---|---|---|---|
| He | $1s^2$ | 99 | **2372 (Highest in PT!)** | +48 | **4.2 (Lowest in PT!)** |
| Ne | [He] $2s^2\ 2p^6$ | 160 | 2080 | +116 | 27.1 |
| Ar | [Ne] $3s^2\ 3p^6$ | 190 | 1520 | +96 | 87.2 |
| Kr | [Ar] $3d^{10}\ 4s^2\ 4p^6$ | 200 | 1351 | +96 | 119.7 |
| Xe | [Kr] $4d^{10}\ 5s^2\ 5p^6$ | 220 | 1170 | +77 | 165.0 |
| Rn | [Xe] $4f^{14}\ 5d^{10}\ 6s^2\ 6p^6$ | 230 | 1037 | +68 | 211.0 |

Table 18.1 — Physical and atomic properties of Noble Gases.

KEY RESULT
Helium Records in Chemistry:
- **Highest Ionization Enthalpy** in the entire periodic table (2372 kJ/mol).
- **Lowest Boiling Point** of any known substance (4.2 K at 1 atm).
- Does not solidify even at absolute zero (0 K) under atmospheric pressure; requires ≈25 atm pressure!

## 2. Liquid Helium & Superfluidity (Helium II)

Liquid helium undergoes a second-order phase transition at **2.18 K (the λ-point)** from normal liquid Helium I to **Helium II (Superfluid)**:
- **Zero Viscosity:** Flows through sub-microscopic capillaries without friction; creeps up and over container walls in a thin Rollin film.
- **Enormous Thermal Conductivity:** ≈800 times higher than copper! Liquid Helium II does not boil with bubbles; evaporation occurs silently from the surface.

## 3. Clathrate (Cage / Inclusion) Compounds

- **Definition:** Non-stoichiometric guest-host inclusion compounds where noble gas atoms (guest) are physically trapped inside the hydrogen-bonded cage cavities of host molecules (quinol / hydroquinone or ice), without any chemical bonds.
- **Conditions for Clathrate Formation:**
  - Host: Quinol (p-dihydroxybenzene) crystallised under pressure of noble gas.
  - **Ar, Kr, Xe form clathrates** because their atomic radii (1.9 to 2.2 Å) fit snugly into the 4.0 Å cavities.
  - **He and Ne do NOT form clathrates!** He and Ne atoms are too small and easily slip through the cavity walls without being trapped.

**Xenon Fluorides (XeFn)** — Bartlett's Discovery, VSEPR Geometries & Hydrolysis

In 1962, Neil Bartlett noticed that $O_2$ reacts with PtF6 to form $[O_2]^+ [PtF6]^-$. Knowing the first IE of Xe (1170 kJ/mol) is almost identical to that of $O_2$ (1175 kJ/mol), he reacted Xe with PtF6 to synthesize the first noble gas compound: **$Xe^+ [PtF6]^-$ (orange-yellow solid)**.

## 1. Controlled Direct Synthesis of Xenon Fluorides

| Compound | Xe : $F_2$ Ratio | Conditions (T, P) | Physical State & Geometry |
|---|---|---|---|
| XeF2 | **2 : 1 (Excess Xe)** | 673 K, 1 bar (Nickel tube) | White solid, sublimes at 298 K; **Linear** ($sp^3d$) |
| XeF4 | **1 : 5 Ratio** | 873 K, 7 bar (Nickel tube) | White crystalline solid; **Square Planar** ($sp^3d^2$) |
| XeF6 | **1 : 20 (Excess $F_2$)** | 573 K, 60-70 bar | Colorless crystals (yellow liquid); **Distorted Octahedral** |

Table 18.2 — Synthetic parameters for Xenon Fluorides (NCERT core).

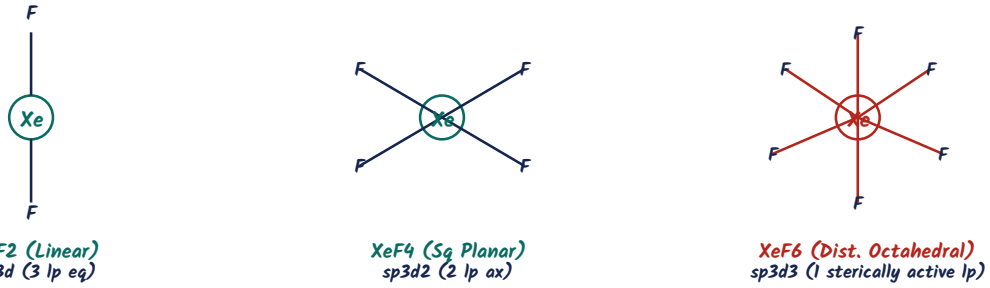



Fig. 18.1 — VSEPR geometries of XeF2 (Linear), XeF4 (Square Planar), XeF6 (Distorted Octahedral).

## 2. Hydrolysis Pathways of Xenon Fluorides (Crucial JEE Topic)

- **1. Hydrolysis of XeF2 (Redox):**
  **$2\ XeF2 + 2\ H_2O \rightarrow 2\ Xe\uparrow + 4\ HF + O_2\uparrow$** (Xe is reduced +2 → 0; O is oxidised -2 → 0).
- **2. Hydrolysis of XeF4 (Disproportionation):**
  **$6\ XeF4 + 12\ H_2O \rightarrow 2\ XeO3\ (+6) + 4\ Xe\ (0) + 24\ HF + 3\ O_2\uparrow$**
  (Xe(+4) disproportionates into Xe(+6) in XeO3 and Xe(0) gas!).
- **3. Hydrolysis of XeF6 (Non-Redox, Stepwise):**
  · Partial hydrolysis (1 eq $H_2O$): **$XeF6 + H_2O \rightarrow XeOF4$ (colorless liquid) + 2 HF**
  · Partial hydrolysis (2 eq $H_2O$): **$XeF6 + 2\ H_2O \rightarrow XeO2F2$ (colorless solid) + 4 HF**
  · Complete hydrolysis (3 eq $H_2O$): **$XeF6 + 3\ H_2O \rightarrow XeO3$ (explosive solid) + 6 HF** (All retain Xe in +6 state!).

TRAP
TRAP: Hydrolysis of XeF4 is a DISPROPORTIONATION reaction that evolves Xe gas, whereas complete hydrolysis of XeF6 is NON-REDOX and produces explosive solid XeO3 without liberating Xe gas!

**Xenon Oxides & Oxofluorides** — VSEPR Geometries, Hybridization & Explosive Nature

Xenon forms explosive oxides XeO3, XeO4, and intermediate oxofluorides XeOF4, XeO2F2, XeOF2.

## 1. Comprehensive Structural Matrix of Xenon-Oxygen Compounds

| Compound | Xe Oxid. State | Hybridization | Lone Pairs | VSEPR Shape | Bond Parameters & Angles |
|---|---|---|---|---|---|
| XeO3 | +6 | $sp^3$ | 1 lone pair | **Pyramidal** (trigonal) | Three Xe=O bonds (dπ-pπ), angle 103° |
| XeO4 | +8 | $sp^3$ | 0 lone pairs | **Regular Tetrahedral** | Four Xe=O double bonds, angle 109.5° |
| XeOF4 | +6 | $sp^3d^2$ | 1 lone pair (trans to =O) | **Square Pyramidal** | Four Xe-F basal + one Xe=O apical |
| XeO2F2 | +6 | $sp^3d$ | 1 equatorial lp | **See-Saw** | Two equatorial Xe=O + two axial Xe-F |
| XeO3F2 | +8 | $sp^3d$ | 0 lone pairs | **Trigonal Bipyramidal** | Three equatorial Xe=O + two axial Xe-F |

Table 18.3 — Complete VSEPR geometries of Xenon Oxides and Oxofluorides.

KEY RESULT
Pi Bonding in Xenon Oxides:
All Xe=O double bonds in xenon chemistry are formed by **dπ-pπ overlap** (donation of oxygen 2p electrons into empty xenon 5d orbitals). For example, XeO3 has 3 dπ-pπ bonds, and XeO4 has 4 dπ-pπ bonds!

## 2. Xenon Trioxide (XeO3) & Xenates

- **Properties of XeO3:** Colorless, hygroscopic, highly explosive crystalline solid. Detonates on touch or heating: $2\ XeO_3(s) \xrightarrow{\text{(shock)}} 2\ Xe(g) + 3\ O_2(g)$ ($\Delta H° = -402$ kJ/mol, violent explosion!).
- **Reaction with Base (Xenate Ion):** XeO3 dissolves in aqueous base to form xenate ion:
  $XeO_3 + OH^- \rightleftharpoons$ **$[HXeO_4]^-$ (Hydrogen xenate ion)**.
  In basic solution, xenate disproportionates slowly into **Perxenate $[XeO_6]^{4-}$** and Xenon gas:
  $2\ [HXeO_4]^- + 2\ OH^- \rightarrow$ **$[XeO_6]^{4-}$ (Perxenate, +8 state) + Xe + $O_2$ + 2 $H_2O$**.
- **Barium Perxenate $Ba_2XeO_6$:** Stable salt. Reacts with cold conc. $H_2SO_4$ to yield volatile, highly explosive gaseous **XeO4 (Xenon tetroxide)**:
  $Ba_2XeO_6 + 2\ H_2SO_4 \rightarrow 2\ BaSO_4\downarrow +$ **$XeO_4\uparrow$** $+ 2\ H_2O$.

**Xenon Fluoride Complexes** — Fluoride Ion Acceptors vs Donors

Xenon fluorides react with fluoride ion acceptors (Lewis acids) to form fluoro-cations, and with fluoride ion donors (alkali metal fluorides) to form fluoro-anions.

## 1. Fluoride Ion Donor Reactions (Forming Xenon Cations)

With strong covalent Lewis acids (MFS where M = P, As, Sb, Bi, Ru, Pt, Nb, Ta), XeFn acts as a **fluoride ion donor** [$F^-$ donor]:

- · $XeF_2 + PFS \rightarrow$ **$[XeF]^+$ $[PF_6]^-$** ($[XeF]^+$ is linear)
- · $XeF_2 + 2\ SbFS \rightarrow$ **$[XeF]^+$ $[Sb_2F_{11}]^-$**
- · $XeF_4 + SbFS \rightarrow$ **$[XeF_3]^+$ $[SbF_6]^-$** ($[XeF_3]^+$ is T-shaped, $sp^3d$ with 2 lone pairs)
- · $XeF_6 + AsFS \rightarrow$ **$[XeFS]^+$ $[AsF_6]^-$** ($[XeFS]^+$ is square pyramidal, $sp^{3d}2$ with 1 lone pair)

| Xenon Fluoride | Lewis Acid / Base | Product Salt Formed | Role of Xenon Fluoride |
|---|---|---|---|
| $XeF_2$ | PFS / SbFS | $[XeF]^+ [PF_6]^-$ / $[XeF]^+ [SbF_6]^-$ | **$F^-$ Donor (Lewis Base)** |
| $XeF_4$ | SbFS | $[XeF_3]^+ [SbF_6]^-$ | **$F^-$ Donor (Lewis Base)** |
| $XeF_6$ | AsFS / SbFS | $[XeFS]^+ [AsF_6]^-$ / $[XeFS]^+ [SbF_6]^-$ | **$F^-$ Donor (Lewis Base)** |
| $XeF_6$ | CsF / RbF (alkali fluorides) | $Cs^+ [XeF_7]^-$ / $Cs_2[XeF_8]$ | **$F^-$ Acceptor (Lewis Acid!)** |

Table 18.4 — Dual Lewis acid-base behaviour of Xenon Fluorides.

KEY RESULT
Dual Nature of $XeF_6$:
$XeF_6$ is amphoteric in the Lewis sense: it can donate $F^-$ to SbFS to form **$[XeFS]^+$**, and it can accept $F^-$ from alkali fluorides MF to form **$[XeF_7]^-$** or **$[XeF_8]^{2-}$** (Octafluoroxenate(VI) ion)!

## 2. Geometry of $[XeF_8]^{2-}$ Anion

In $Cs_2[XeF_8]$ and $NO_2[XeF_8]$, the $[XeF_8]^{2-}$ anion has 8 bond pairs and 1 lone pair. Its structural geometry is a **Square Antiprismatic** arrangement ($D_{4d}$ symmetry). It is thermally stable up to 400°C.

**Krypton, Radon & JEE Applications** — KrF2, Radon Chemistry & Industrial Uses

Beyond Xenon, chemical reactivity in Group 18 is severely limited by ionization energy.

## 1. Chemistry of Krypton & Radon

- **Krypton Difluoride (KrF2):** The only well-characterised krypton compound. Prepared by electrical discharge or UV irradiation of Kr + $F_2$ at -196°C:
  Kr + $F_2$ —(glow discharge, 77 K)→ **KrF2 (colorless volatile solid)**.
  Linear structure ($sp^3d$, 3 equatorial lone pairs). Thermally unstable (decomposes at 0°C). It is an even **stronger fluorinating agent than XeF2**:
  Ag + KrF2 → AgF2; Au + KrF2 + SbF5 → **$[AuF_6]^-$** (oxidises Gold to Au(+V)!).
- **Radon Chemistry:** Radon is radioactive (Rn-222, $t_{1/2}$ = 3.82 days, α-emitter). Forms **RnF2** (identified by radiochemical tracer and electromigration techniques). Covalent chemistry is difficult to study due to intense radiolysis.
- **Helium, Neon, Argon:** No neutral isolable chemical compounds known at standard conditions. Molecular ions like $HeH^+$, $He_2^+$, $ArH^+$ detected only in mass spectrometers.

## 2. Master Summary of Noble Gas Uses in JEE

| Gas | Physical Property Utilised | Industrial, Laboratory & Medical Applications |
|---|---|---|
| Helium (He) | Low density & non-inflammable | Filling weather balloons and airships (safe substitute for $H_2$) |
| Helium (He) | Low solubility in blood at high pressure | Deep-sea diving gas cylinders (mixture of **80% He + 20% $O_2$**) to prevent the painful decompression sickness ('the bends') |
| Helium (He) | Extremely low boiling point (4.2 K) | Cryogenic coolant for superconducting magnets in **MRI (Magnetic Resonance Imaging)** and NMR |
| Neon (Ne) | Bright orange-red glow in discharge tubes | **Neon advertising display signs** and beacon lights for airfield navigation (penetrates fog) |
| Argon (Ar) | Complete chemical inertness & cheap | Inert gas atmosphere in metallurgical arc-welding of Al and stainless steel; filling incandescent light bulbs |
| Xenon & Krypton | Intense bright flash on discharge | High-speed photography strobe flashbulbs; lasers |
| Radon (Rn) | Radioactive α-emission | Radiotherapy for cancer treatment; hydrological tracing |

Table 18.5 — Practical applications of Noble Gases in JEE.

**Master Cross-Group Review** — Comparative Periodic Trends Across Groups 13 to 18

A consolidated comparison of anomalous trends, inert pair redox patterns, and Lewis acidity across all six p-block groups.

## 1. Atomic Radii & Ionization Energy Anomalies Summary

| Property / Trend | Standard Group Expectation | Actual Observed Sequence & Underlying Physical Reason |
|---|---|---|
| Atomic Radius in G13 | B < Al < Ga < In < Tl | **B < Ga < Al < In ≈ Tl** (Ga < Al due to 3d transition contraction, $d^{10}$ poor shielding) |
| Ionization Enthalpy in G13 | B > Al > Ga > In > Tl | **B > Tl > Ga > Al > In** (Tl > In due to 4f lanthanoid contraction, high $Z_{eff}$) |
| Ionization Enthalpy in G14 | C > Si > Ge > Sn > Pb | **C > Si > Ge > Pb > Sn** (Pb > Sn due to 4f contraction) |
| IE-1: G15 vs G16 | Group 16 > Group 15 | **Group 15 > Group 16** (N > O, P > S, As > Se due to stable half-filled $np^3$ configuration) |
| Electron Gain Enthalpy (G16) | O > S > Se > Te | **S > Se > Te > Po > O** (Oxygen is LEAST negative due to compact 2p inter-electronic repulsions) |
| Electron Gain Enthalpy (G17) | F > Cl > Br > I | **Cl > F > Br > I** (Chlorine is MOST negative in entire periodic table; F has compact 2p repulsions) |
| Bond Dissociation Energy (G17) | $F_2 > Cl_2 > Br_2 > I_2$ | **$Cl_2 > Br_2 > F_2 > I_2$** ($F_2$ has low bond energy due to severe lone-pair repulsions, 143 pm) |

Table R.1 — Master compendium of p-block periodic anomalies.

## 2. The Inert Pair Effect Matrix Across p-Block

| Group | Lower State (+n) | Higher State (+n+2) | Oxidising Agent (+n+2 State) | Reducing Agent (+n State) |
|---|---|---|---|---|
| Group 13 | +1 ($Tl^{+}$ stable) | +3 ($B^{3+}$, $Al^{3+}$ stable) | **$Tl^{3+}$ is strongly oxidising** ($Tl^{3+} + 2e^{-} \rightarrow Tl^{+}$) | $Al^{+}$, $Ga^{+}$ (disproportionate) |
| Group 14 | +2 ($Pb^{2+}$ stable) | +4 ($C^{4+}$, $Si^{4+}$ stable) | **$Pb^{4+}$ is strongly oxidising** ($PbO2 + 4\ HCl \rightarrow PbCl2 + Cl_2$) | **$Sn^{2+}$ is strongly reducing** ($Sn^{2+} \rightarrow Sn^{4+} + 2e^{-}$) |
| Group 15 | +3 ($Bi^{3+}$ stable) | +5 ($N^{5+}$, $P^{5+}$ stable) | **$Bi^{5+}$ is strongly oxidising** ($NaBiO3$ oxidises $Mn^{2+}$ to $MnO4^{-}$) | $P^{3+}$, $As^{3+}$ (mild reducing) |

Table R.2 — Comparative redox chemistry governed by the inert pair effect.

KEY RESULT

Inert Pair Generalisation:

· The lower oxidation state becomes progressively **MORE stable** descending the group: Tl(+) > In(+) > Ga(+) > Al(+) > B(+).

· The higher oxidation state becomes progressively **A STRONGER OXIDANT** descending the group: Tl(3+) > In(3+) > Ga(3+) > Al(3+).

**Complete Hydrolysis Map** — Mechanisms, Redox vs Non-Redox & Intermediate Acids

Hydrolysis in the p-block occurs via two distinct pathways: nucleophilic attack on empty d orbitals (SN2-type), or coordination to an empty p orbital (Lewis acid-base). Below is the comprehensive master reaction map.

## 1. Non-Redox Hydrolysis (Oxidation State Preserved)

| Substrate | Hydrolysis Equation | Nature of Attack / Intermediate |
|---|---|---|
| $BCl_3$, $BBr_3$, $BI_3$ | $BX_3 + 3\ H_2O \rightarrow$ **$H_3BO_3 + 3\ HX$** | Water attacks vacant 2p orbital on Boron |
| $BF_3$ | **$4\ BF_3 + 3\ H_2O \rightarrow H_3BO_3 + 3\ H[BF_4]$** | HF formed coordinates with unreacted $BF_3$ to give fluoroboric acid! |
| $SiCl_4$ | $SiCl_4 + 4\ H_2O \rightarrow$ **$Si(OH)_4$ (Silicic acid) → $SiO_2{\cdot}2H_2O + 4\ HCl$** | Attack on vacant 3d orbitals; coordination number expands to 6 |
| $PCl_3$ | $PCl_3 + 3\ H_2O \rightarrow$ **$H_3PO_3$ (Phosphorous acid) + 3 HCl** | Preserves +3 state; lone pair remains on P |
| $PCl_5$ | $PCl_5 + H_2O \rightarrow$ **$POCl_3$** + 2 HCl; $POCl_3 + 3\ H_2O \rightarrow$ **$H_3PO_4 + 3\ HCl$** | Stepwise via phosphorus oxychloride |
| $SF_4$ | $SF_4 + 2\ H_2O \rightarrow$ **$SO_2 + 4\ HF$** | Water attacks vacant d-orbitals; preserves +4 state |
| $XeF_6$ | $XeF_6 + H_2O \rightarrow$ **$XeOF_4$** + 2 HF → **$XeO_2F_2$** + 4 HF → **$XeO_3 + 6\ HF$** | Stepwise replacement of $F_2$ by =O; all preserve Xe(+6) |

Table R.3 — Master non-redox hydrolysis reactions.

## 2. Redox & Disproportionation Hydrolysis Reactions

| Substrate | Reaction Equation | Type of Redox Reaction |
|---|---|---|
| $NCl_3$ | $NCl_3 + 3\ H_2O \rightarrow$ **$NH_3$ (-3) + 3 HOCl (+1)** | Non-redox on paper, but N acts as negative end ($\chi_N > \chi_{Cl}$)! Cl goes to HOCl, NOT HCl! |
| $P_4$ (in alkali) | $P_4 + 3\ NaOH + 3\ H_2O \rightarrow$ **$PH_3$ (-3) + 3 $NaH_2PO_2$ (+1)** | Disproportionation of White Phosphorus |
| $Cl_2$ (in water) | $Cl_2 + H_2O \rightleftharpoons$ **HCl (-1) + HOCl (+1)** | Disproportionation (forms chlorine water; turns blue litmus red then bleaches) |
| $XeF_2$ | **$2\ XeF_2 + 2\ H_2O \rightarrow 2\ Xe\ (0) + 4\ HF + O_2\ (0)$** | **Redox:** Xe(+2) reduced to 0; $H_2O$(-2) oxidised to $O_2$(0) |
| $XeF_4$ | **$6\ XeF_4 + 12\ H_2O \rightarrow 2\ XeO_3$ (+6) + 4 Xe (0) + 24 HF + $3\ O_2\uparrow$** | **Disproportionation + Redox:** Xe(+4) splits into +6 and 0 |

Table R.4 — Redox and disproportionation hydrolysis pathways in JEE.

KEY RESULT

The Classic $NCl_3$ vs $NF_3$ vs $PCl_3$ Hydrolysis Trap:

- **$NF_3$:** Does NOT hydrolyse at all (no vacant d-orbitals, high bond strength, non-polar).
- **$NCl_3$:** Hydrolyses to **$NH_3 + HOCl$** (because $\chi_N = 3.0 > \chi_{Cl} = 2.8$; water attacks Cl with empty 3d orbitals!).
- **$PCl_3$:** Hydrolyses to **$H_3PO_3 + HCl$** (because $\chi_P = 2.1 < \chi_{Cl} = 3.0$; water attacks P with empty 3d orbitals!).